Received: 28 April 2023 | Accepted: 10 February 2025

DOI: 10.1111/1911-3846.13042

ARTICLE

CAAA ACPC CONTEMPORARY ACCOUNTING RESEARCH | RECHERCHE COMPTABLE CONTEMPORAINE

Does tax enforcement inform auditors' risk assessment? Evidence from key audit matters

Jessica R. Filosa[1] | **Jingjing (Jing) Huang**[2] | **Lijun (Gillian) Lei**[3] | **Sarah E. Stein**[2]

[1]Culverhouse School of Accountancy, University of Alabama, Tuscaloosa, Alabama, USA

[2]Pamplin College of Business, Virginia Tech, Blacksburg, Virginia, USA

[3]Bryan School of Business and Economics, University of North Carolina, Greensboro, North Carolina, USA

Correspondence
Sarah E. Stein, Pamplin College of Business, Virginia Tech, Blacksburg, VA, USA.
Email: sestein@vt.edu

Abstract

International standards encourage auditors to consider regulatory factors and external parties during the risk assessment process. One such external party is the taxation authority, which monitors corporate conduct and uses the threat of tax audits to constrain managerial opportunism. This study examines whether tax enforcement influences auditors' perception of the risk of material misstatement on their engagements. Using a sample of companies listed on European exchanges, we assess the strength of tax enforcement at the country level based on the number of full-time equivalent (FTE) employees in the tax audit and verification function relative to the size of the economy. Since auditors of these European listed companies must publicly disclose key audit matters (KAMs), we use the number of KAMs to capture auditors' perceptions of client-level misstatement risk. Our results indicate that auditors report fewer KAMs in the presence of more FTEs in the tax audit and verification function. In cross-sectional tests, we find that this negative association is stronger in settings where auditors are more inclined to incorporate the monitoring potential of the tax authority into their risk assessment, such as in countries with high book-tax conformity and for auditors with greater exposure to complex tax issues. These findings offer new insights into the role of tax enforcement in auditors' decision-making and have timely implications for accounting regulators and academics studying the determinants of KAMs.

KEYWORDS

key audit matters, monitoring, risk assessment, tax enforcement

Accepted by Dechun Wang.

© 2025 Canadian Academic Accounting Association.

La mise en recouvrement de l'impôt influence-t-elle l'évaluation des risques par les auditeurs? Données issues des questions clés de l'audit

Résumé

Les normes internationales encouragent les auditeurs à prendre en compte les facteurs réglementaires et les parties externes lors du processus d'évaluation des risques. L'une de ces parties externes est l'administration fiscale, qui contrôle la conduite des entreprises et utilise la menace de contrôles fiscaux pour limiter l'opportunisme des dirigeants. Cette étude examine si la mise en recouvrement de l'impôt influence la perception qu'ont les auditeurs du risque d'anomalies significatives dans le cadre de leurs missions. À partir d'un échantillon d'entreprises cotées sur les bourses européennes, nous évaluons la force de la mise en recouvrement de l'impôt au niveau national en fonction du nombre d'employés équivalents temps plein (ETP) dans la fonction d'audit et de vérification fiscale par rapport à la taille de l'économie. Étant donné que les auditeurs de ces sociétés européennes cotées en bourse doivent divulguer publiquement les questions clés de l'audit (QCA), nous utilisons le nombre de QCA pour saisir les perceptions des auditeurs quant au risque d'anomalies au niveau du client. Nos résultats indiquent que les auditeurs divulguent moins de QCA en présence d'un plus grand nombre d'ETP dans la fonction de contrôle et de vérification fiscale. Dans les tests transversaux, nous constatons que ce lien négatif est plus fort dans les contextes où les auditeurs sont plus enclins à intégrer le potentiel de contrôle de l'administration fiscale dans leur évaluation des risques, comme dans les pays où la conformité entre les comptes et les impôts est élevée et pour les auditeurs plus exposés à des questions fiscales complexes. Ces résultats offrent de nouvelles perspectives sur le rôle de l'application de la législation fiscale dans la prise de décision des auditeurs et ont des implications opportunes pour les autorités de réglementation de la comptabilité et les universitaires qui étudient les déterminants des questions clés de l'audit.

MOTS-CLÉS

application de la législation fiscale, contrôle, évaluation des risques, questions clés de l'audit

JEL CLASSIFICATION

M41, M42, M48, H25, H26.

1 | INTRODUCTION

International standard setters emphasize the importance of the auditor's risk assessment process, which sets the tone for the nature, timing, and extent of audit procedures to be performed on each client engagement (IAASB, 2020a, 2020b; PCAOB, 2017a, 2017b). When assessing potential misstatement risks, international standards encourage auditors to consider regulatory factors and other external parties that review and analyze the company's financial performance (IAASB, 2020b, paras. 19, A70, and A79). One of the external parties explicitly highlighted in the standards is the taxation authority. Although their purpose is not to audit financial reporting information, tax authorities have access to proprietary company data and can act as a complementary monitor through their enforcement actions. As such, the enforcement efforts of tax authorities may inform auditors' risk assessment and influence their perceptions of companies' risk of material misstatement (RMM). Therefore, this study investigates whether country-level tax enforcement is associated with auditors' assessment of client-level misstatement risk.

Due to their tax collection function, tax authorities have strong incentives to monitor managers in order to protect the government's claims on corporate cash flows (Desai et al., 2007; Dyck & Zingales, 2004) and reduce managers' incentive to hoard or conceal bad news (Bauer et al., 2021; Guedhami & Pittman, 2008; Yost & Shu, 2022). Prior research documents that tax enforcement influences managers' perception of ex ante tax audit likelihood and constrains their opportunistic behaviors, resulting in less tax avoidance (De Simone et al., 2023; Hoopes et al., 2012), better financial reporting quality (Hanlon et al., 2014), lower stock price crash risk (Bauer et al., 2021), and improved creditworthiness of borrowers (Gallemore & Jacob, 2020). While these studies examine changes in companies' financial outcomes due to the threat of tax enforcement, our focus is on whether external auditors incorporate the potential for tax enforcement into their perception of misstatement risk on their engagements. Specifically, our focus is on auditors' consideration of the ex ante potential for tax enforcement rather than their observation of these ex post corporate financial outcomes. If auditors perceive that the tax authority's monitoring efforts will curb managerial opportunism, then we predict that auditors will adjust their risk assessment downward in the presence of strong tax enforcement.

We rely on data from the International Survey on Revenue Administration (ISORA) to determine the strength of tax enforcement. This tax survey provides internationally comparative data on various aspects of tax enforcement, including the number of full-time equivalent (FTE) employees focusing on the tax audit and verification function. This function within the tax authority performs one of its most important enforcement tasks (Blaufus et al., 2022; Nessa et al., 2020). As part of this function, tax authorities inspect a vast number of records during field audits and use third-party information to verify tax return data and enhance their enforcement efforts (Deloitte, 2021; OECD, 2017a, 2021). We argue that the verification efforts of the tax authority help constrain corporate misbehavior and set expectations for higher quality reporting, thereby generating externalities that may affect auditors' perception of misstatement risk. Motivated by prior research that examines tax enforcement by using tax audit probability (e.g., Bauer et al., 2021; Guedhami & Pittman, 2008), we measure the strength of tax enforcement using the ISORA data to capture the likelihood of tax audits in the economy. Specifically, we scale the number of FTEs in the tax audit and verification function by inflation-adjusted gross domestic product (GDP) in each country and year.[1]

[1]Relative to those prior studies in the United States, we lack public data on the tax authority's audit rate for a broad sample of companies from different countries. Therefore, we measure the likelihood of tax audits in the economy as noted above to capture the perceived tax audit risk by external stakeholders, such as financial statement auditors. For robustness, we find similar evidence when using the tax enforcement budget scaled by GDP (De Simone et al., 2023). Nevertheless, the budget data cover various functions not directly related to the tax authority's audit and verification function, such as administrative support and retiree pensions. These administrative functions introduce measurement noise that complicates our ability to accurately capture auditors' perceptions of the strength of tax enforcement.

To discern auditors' perception of client-level misstatement risk, we rely on a recent change to the auditor's report that requires disclosure of risk-related matters to financial statement users (Minutti-Meza, 2021). Specifically, International Standard on Auditing (ISA) 701 states that auditors must disclose key audit matters (KAMs), defined as "those matters that, in the auditor's professional judgment, were of *most significance* [emphasis added] in the audit of the financial statements of the current period" (IAASB, 2020c, para. 8). KAMs do not provide additional assurance or qualify the audit opinion; their purpose is solely to provide greater transparency into the audit. Since KAMs reflect areas of substantial risk and judgment, they offer insights into the auditor's risk assessment process by revealing which and how many areas the auditor determined to be of most significance during the engagement (Rousseau & Zehms, 2024; K. W. Smith, 2023). For example, Lynch et al. (2024) show that KAMs can provide a confirmatory risk signal to stakeholders. They document that investors lower their valuation of tax avoidance and deferred tax assets when auditors issue a tax-specific KAM. Similarly, Klevak et al. (2023) report that both investors and analysts perceive critical audit matters (CAMs)—the US equivalent of KAMs—as a signal of overall business risk. Given the capacity for KAMs to signal risk, we use the number of KAMs to capture auditors' assessment of the RMM.[2]

Using a sample of companies listed on European stock exchanges between 2014 and 2019, we find that auditors perceive less RMM in the presence of strong tax enforcement. That is, auditors report fewer total KAMs when there is a higher likelihood of a tax audit (i.e., more FTEs working in the tax authority's audit and verification function) after controlling for other factors such as client complexity, financial reporting quality, tax avoidance, audit fees, auditor type, and country, industry, and year fixed effects. In additional robustness tests, we demonstrate that our core results hold when including company and year fixed effects, using a change specification, developing an instrumental variable approach, employing entropy balancing techniques, and using alternative measures. Since our sample includes auditors' risk disclosures reported under two different regimes, we also document that the results are robust to removing observations from the United Kingdom under the prior standard as well as using a subsample of firms disclosing KAMs for the first time.

While the primary findings have intuitive appeal, it is less certain whether the effect of strong tax enforcement is confined to companies' tax accounts or if it extends more broadly to other financial statement accounts. After disaggregating KAMs into tax and nontax categories following the topic definitions provided by Audit Analytics, we find that the increased likelihood of tax audits is associated with fewer tax *and* nontax KAMs. Thus, the externalities generated by the tax authority through its enforcement efforts appear to influence auditors' perception of risk in both tax and nontax areas of the audit engagement. This evidence complements the findings in Chow et al. (2022), who demonstrate that tax enforcement intensity in China is associated with fewer tax-related misstatements and other types of accounting misstatements.

To provide further context and support for our main findings, we develop two cross-sectional predictions whereby the potential for tax enforcement is more likely to influence auditors' perception of misstatement risk. We begin by focusing on a country-level factor—book-tax conformity—where auditors can more readily leverage tax authorities' verification efforts since book and tax reporting requirements are more comparable. For example, when the two reporting systems require similar recognition of expenses, it is difficult for a company to opportunistically report a book expense that substantially differs from the amount recorded for tax purposes. Empirically, we construct a measure of book-tax conformity for each country-year,

[2]Though we predict a negative association between the number of KAMs and the likelihood of tax audits, some studies suggest that KAMs do not provide incremental information or useful insights into the audit process (e.g., Gutierrez et al., 2018; Lennox et al., 2023). If variation in KAM disclosures does not reflect auditors' assessment of client-level misstatement risk, then we will fail to find evidence supporting our prediction.

following Atwood et al. (2010). Our results indicate that the negative association between the likelihood of tax audits and the number of KAMs is stronger in settings with high book-tax conformity. When we separately analyze tax versus nontax KAMs, we find that the documented effect related to high book-tax conformity is concentrated in nontax KAMs. This pattern of evidence suggests that closer alignment between book and tax reporting influences whether auditors consider the potential for tax enforcement when assessing nontax misstatement risk. In contrast, book-tax conformity appears to have less impact on the relation between tax enforcement and the auditors' perception of risk in tax-specific accounts.

Next, we examine an auditor-specific factor—exposure to complex tax issues within their client base—that may influence how auditors consider the potential for tax enforcement in their risk assessment. When external auditors have high exposure to complex tax issues that attract the scrutiny of tax authorities, we expect them to be more familiar with the tax authority's enforcement efforts and, therefore, more readily able to incorporate the threat of tax audits into their perception of misstatement risk. Motivated by Goldman et al. (2022), we focus on whether the auditor's local clientele exhibit characteristics that are commonly associated with complex tax issues, such as R&D, multinational activity, and mergers and acquisitions (M&A). Our results indicate that the negative association documented in our primary tests is stronger when auditors have high exposure to complex tax issues. When we separately analyze tax and nontax KAMs, this result appears to be concentrated in nontax KAMs. This evidence suggests that auditors may place greater emphasis on the potential monitoring strength of the tax authority when assessing risk for nontax accounts. In contrast, as auditors accumulate tax-specific knowledge from working with clients that face complex tax issues, they might rely on their own knowledge instead of the perceived strength of tax enforcement when assessing the risk of tax accounts.[3]

In a final test, we evaluate whether other institutional factors within each country could be driving our results. Specifically, we follow ISA 315 and construct variables capturing other parties within the external environment (e.g., analyst following, media, other regulators, trade unions, and providers of finance) (IAASB, 2020b). We find that our primary results continue to hold after controlling for these external parties. Importantly, the variables capturing these other external parties are not highly correlated with the likelihood of tax audits, suggesting our measure of tax enforcement is distinct.

Overall, our study contributes to existing research and practice in several ways. Prior research focuses on managers' response to tax enforcement such as changes in tax avoidance and earnings management (De Simone et al., 2023; Hanlon et al., 2014; Nessa et al., 2020). We extend this research by providing new evidence on the externality of tax enforcement for financial statement auditors.[4] In a related concurrent study based on a tax administration change in China, Chow et al. (2022) document that auditors exert less effort and simultaneously improve audit quality when their clients are monitored by a more stringent tax regime. Differing from their study, we analyze an international sample as opposed to a single country. This approach allows us to provide evidence on a specific channel through which tax enforcement influences auditors' perception of misstatement risk—that is, variation in book-tax conformity across countries. Moreover, we show that the externality of tax enforcement is particularly strong when auditors have high exposure to complex tax issues. The collective findings from

[3]In both sets of subsample tests (book-tax conformity and auditor exposure to complex tax issues), it is difficult to rule out that the lack of power—due to the low frequency of tax KAMs—may be driving these insignificant results. Therefore, we caution that the findings should be interpreted with this caveat in mind.

[4]For example, Hanlon et al. (2014) find a decrease in accrual-based earnings management when there is a higher probability of an IRS audit, which supports the monitoring role of tax enforcement in constraining opportunistic behaviors. In contrast, our focus is on external auditors' consideration of the ex ante potential for tax enforcement rather than their observation of these ex post corporate financial outcomes. Moreover, factors beyond the client's discretionary accruals quality influence the identification of KAMs (e.g., accounts based on complex estimates such as goodwill, issues with information technology systems, and so on) such that it is not obvious that lower levels of discretionary accruals would explain auditors reporting fewer KAMs.

Chow et al. (2022) and our study bring new evidence to the role of tax enforcement in auditors' decision-making.

Our study also extends research examining the extent to which country-level factors and other external parties influence auditors' decisions. This research demonstrates that auditors respond to cross-country differences in legal and investor protection regimes (Choi et al., 2008; Francis & Wang, 2008) and suggests that auditors consider other types of outside parties in their risk assessment. Examples of those parties examined in prior research include short sellers (Hope et al., 2017), institutional owners (Cassell et al., 2018, 2022; Lim et al., 2013), analysts (Gotti et al., 2012), and regulators (Boo & Sharma, 2008; Lamoreaux, 2016). Building on this research, our results suggest that auditors respond to cross-country differences in tax enforcement by lowering their perceptions of risk in countries with a heightened likelihood of tax audits. Moreover, the effect of country-level tax enforcement on auditors' risk assessment is incremental to other external factors cited in the auditing standards.

We also contribute to the stream of research investigating the external auditors' risk assessment process. Although this process is pivotal to the execution of the audit, the traditional audit report does not allow external stakeholders to understand the greatest RMMs in the eyes of the auditor. Due to this data limitation, prior research often uses audit fees—or in some cases, audit hours or labor mix—to infer auditors' perceptions of client risk (Bae et al., 2021; Bell et al., 2008; Cho et al., 2017; Hogan & Wilkins, 2008; O'Keefe et al., 1994). Instead, we add to recent research using KAMs as an alternative approach to capture auditors' perception of risk (Burke et al., 2023; Sierra-Garcia et al., 2019). Our use of KAMs for this purpose responds to a call from DeFond and Zhang (2014) to use "creative settings and research designs to open up the 'black box' [of the audit process]" (p. 278). Further, Christensen et al. (2019) highlight the inability of traditional audit reports to fully signal risk as auditors observe it, and they call on future research to examine whether "critical audit matter paragraphs fill the information gap" (p. 1490). We argue that KAMs provide a unique opportunity to analyze auditors' professional judgment in an international setting since auditors must determine and disclose which matters have the *most* significant influence on the engagement.

Finally, the evidence in this study adds to our understanding of the determinants of KAM disclosures. The IAASB and PCAOB are currently undertaking post-implementation reviews of the changes to auditor reporting standards (IAASB, 2020d; PCAOB, 2022), and the goal is to gauge whether the standards are implemented in a way that achieves their intended purpose. Developing a robust understanding of the factors considered in KAM reporting decisions will lead to a better appreciation for the information that KAMs convey, and thus, whether and how various stakeholders should use this information. While prior studies on the determinants of KAMs primarily examine client- or auditor-specific characteristics (e.g., Burke et al., 2023; Lynch et al., 2024; Pinto & Morais, 2019; Sierra-Garcia et al., 2019), we broaden the literature by considering a country-level factor. Our study may inform regulators about the extent to which aspects of the institutional environment inform auditors' identification and disclosure of KAMs, which sheds light on one aspect of how the standard has been implemented thus far.

2 | BACKGROUND AND EMPIRICAL PREDICTIONS

2.1 | Institutional details regarding tax enforcement

In recent years, European countries have been working with the OECD to ramp up their tax enforcement efforts. For example, the OECD and G20 countries pledged resources to strengthen the data transmission systems of tax authorities and to increase the exchange of information across tax administrations in different countries (Forum on Tax Administration, 2014, 2016). Generally speaking, the tax authority allocates resources across distinct functions including tax audit and verification; registration, processing, and services; debt management; support; and other operations (OECD, 2017a). Tax audit and verification is the "dominant administration function" as it conducts one of the tax authority's most important enforcement tasks and, as such, receives the most staff resources on average (OECD, 2017a, p. 36). Beyond collecting tax revenue, tax audit and verification activities are important to "secure the system, influence wider behavior, and reinforce social norms" (OECD, 2017b, p. 30). The employees within this function may "undertake systems audit on data/numbers, interview staff, and make extensive requests for data" (Deloitte, 2021, p. 33). Shedding light on the mechanics of tax enforcement, Blaufus et al. (2022) provide these details about how tax audits are conducted in Germany:

> Tax audits are conducted as field audits. In a field audit, the Revenue Agency conducts a detailed examination of a taxpayer's records, commonly at the taxpayer's place of business. During the audit, auditors usually identify certain items they disagree with concerning the taxpayer's chosen tax treatment. In a final audit meeting, the [tax] auditor discusses face-to-face with the firm (usually represented by a tax advisor) unclear issues including items in which the respective tax treatment is unclear due to tax ambiguity. (pp. 1–2)

As part of their verification efforts, tax auditors can request access to a company's records (e.g., inventory management, fixed assets, and payroll) through various channels, including direct on-site access through the company's information system, assisted on-site access based on specific requests, and off-site access for the tax auditors to use as needed (Vogele & Bader, 2001).

During tax audits, tax authorities use an extensive amount of information to scrutinize tax return information and enhance their enforcement efforts. The OECD (2021) *Tax Administration* report highlights various data sources that tax authorities access for compliance purposes, including (1) transactional data collected from devices such as online cash registers; (2) data from banks or payment intermediaries to allow direct verification of a taxpayer's income or assets; (3) supplier data that are collected directly through e-invoicing systems; (4) unstructured internet and social media information about the taxpayer; and (5) data held by other government agencies related to licensing, regulatory, or social security purposes. Tax authorities can also request taxpayers' business transaction information from financial institutions, international exchanges, and online trading. Following these guidelines, one example is that the UK tax authority can scrutinize daily business transactions of companies, in addition to examining financial records (HM Revenue and Customs, 2016). Using diverse types of third-party information, tax authorities can cross-check tax returns and strengthen their verification efforts, resulting in greater enforcement to deter tax avoidance (Adhikari et al., 2021; Lederman, 2010; Pomeranz, 2015; Slemrod et al., 2017). Overall, tax authorities' access to various internal and third-party data gives them a clear information advantage in assessing business risk and monitoring corporate conduct.

2.2 | External auditors' risk assessment

International auditing standards require auditors to identify and assess the RMM based on an understanding of the client and its environment (IAASB, 2020a, 2020b). Specifically, auditors must identify and assess inherent risks that arise due to the nature and complexity of the company's business—including the macroeconomic environment—before considering controls designed and executed by management to mitigate such risks. Inherent risk is unique to each company and is initially assessed by the auditor during the planning phase of each engagement.

The standards require auditors to consider engagement-specific context and apply professional judgment when assessing inherent risk and developing a response to these risks (Causholli & Knechel, 2012). Prior research highlights that the auditor's risk assessment process is subjective, considers many factors, and requires various types of expertise (Allen et al., 2006; Eilifsen et al., 2001; O'Keefe et al., 1994; Taylor, 2010; Trotman & Wright, 2012).

External circumstances that give rise to business risk may influence the auditor's assessment of inherent risk as noted in paragraphs A25 and A40 of ISA 200 (IAASB, 2020a). Paragraph A79 of ISA 315 specifically cites external parties for the auditor to consider including analysts, news and media, *taxation authorities* [emphasis added], regulators, trade unions, and providers of finance (IAASB, 2020b). Prior research provides insights into how certain external parties impact auditor behavior. For example, studies document that auditors respond to the actions of short sellers (Hope et al., 2017), institutional owners (Cassell et al., 2018, 2022; Lim et al., 2013), analysts (Gotti et al., 2012), and regulators (Boo & Sharma, 2008; Lamoreaux, 2016). Building on this literature, we investigate whether the enforcement efforts of tax authorities inform auditors' risk assessment.

We believe focusing on the role of tax authorities is important for several reasons. Unlike the consistent incentive function of tax authorities, financial regulators and the media are influenced by the volatility of politics and popular sentiment; as a result, their effect on company-level reporting risk is less predictable. Moreover, given that the tax authority uses proprietary tax returns, it is less likely to pursue litigation against the auditor over public financial information relative to other external parties (e.g., analysts, credit agencies, trade unions, debtholders). Thus, our setting differs from prior research on external factors that influence auditor liability if an error is uncovered, such as shareholder litigation and financial enforcement (Ewert & Wagenhofer, 2019; Florou et al., 2020; Sulcaj, 2023).

2.3 | Development of hypotheses

2.3.1 | Primary prediction

Tax authorities' monitoring efforts can help to mitigate various forms of corporate misconduct. For example, research demonstrates that—through the threat of tax audits—tax enforcement helps to reduce corporate tax avoidance (Hoopes et al., 2012). As tax authorities conduct comprehensive tax audits, their data verification efforts can also strengthen the reliability of accounting records and internal reporting systems, thereby constraining managers' opportunistic behaviors and reducing agency costs. Consistent with this reasoning, prior research suggests that tax enforcement improves companies' information environments. For example, tax enforcement intensity is associated with improved financial reporting quality (Hanlon et al., 2014) and creditworthiness (Gallemore & Jacob, 2020). When tax enforcement is strong, investors also increase their valuation of taxable income incremental to book income (Kerr, 2019), demand a lower cost of equity capital (El Ghoul et al., 2011), and face lower stock price crash risk (Bauer et al., 2021). Similarly, tax enforcement benefits debtholders as evidenced through the demand for a smaller risk premium on private bond issuance (Guedhami & Pittman, 2008). Importantly, these studies use the probability of tax audits to capture the potential of tax enforcement, which implies that the threat of tax audits can play a monitoring role in deterring managerial opportunistic behaviors. These findings are further supported by the OECD report, which shows that for companies not subject to a tax audit themselves, the tax authorities' monitoring efforts can increase their perceived power and improve compliance (OECD, 2017b, p. 36).

This collective evidence suggests that the monitoring role of tax enforcement may influence the external auditor's perception of misstatement risk. Given that international standards encourage auditors to consider taxation legislation and regulation as well as public information published by tax authorities during their risk assessment process (IAASB, 2020b), we expect that auditors heed this advice in practice. As such, we predict that auditors will incorporate the monitoring potential of strong tax enforcement and adjust their risk assessments accordingly. Thus, our first prediction is stated in the alternative form as follows:

> **Hypothesis 1 (H1).** External auditors identify less RMM in the presence of strong tax enforcement.

While there are valid reasons to believe tax enforcement informs auditors' risk perceptions, our prediction relies on auditors' indirect assessment of their clients' environment rather than direct communication with the tax authority about their enforcement efforts. Therefore, we may fail to find evidence supporting our prediction if auditors do not observe the efforts of the tax authority or if they view tax enforcement as irrelevant to their risk assessment.

2.3.2 | Cross-sectional predictions

In this section, we focus on two settings where external auditors are more likely to perceive the potential monitoring benefits of tax enforcement: (1) book-tax conformity within the country and (2) the auditor's exposure to complex tax issues.

2.3.2.1 | *Book-tax conformity*

Countries generally have two separate reporting systems—book and tax. Companies report financial accounting income (i.e., book income) to various stakeholders under the applicable reporting standards and simultaneously report taxable income to the tax authority following the tax rules.[5] Although the practical differences between the two reporting regimes can be quite subtle, the two frameworks fundamentally differ in their objectives (Hanlon et al., 2005). The tax rules reflect the government's goal to collect tax revenue and offer tax incentives. In contrast, the objective of financial accounting standards is to offer relevant and reliable information to users in making decisions related to providing resources to the entity. As these two systems diverge, book income moves away from taxable income and book-tax conformity decreases. While it is challenging to directly compare book and tax frameworks to determine the extent of similarity, Atwood et al. (2010) use a novel approach to capture the co-movement of book and taxable income at the country-year level and identify significant variation in book-tax conformity across countries.

We conjecture that the potential monitoring benefits of tax enforcement will extend beyond the tax accounts, particularly when book-tax conformity is high. When tax and financial standards have similar reporting requirements, the verification of taxable income by tax authorities is more likely to benefit financial accounting verification by auditors. For example, when these two systems recognize expenses in a similar way, it is difficult for a company to report a book expense that substantially differs from the amount recorded for tax purposes. Consistent with this assertion, Bakarich and Kerr (2016) find that audit quality is higher in countries with greater book-tax conformity. Thus, we predict that strong tax enforcement will have a greater influence on the auditor's perception of risk in countries with more conformity in their book and tax reporting systems. We state our second hypothesis in the alternative form as follows:

[5]For book purposes, managers have incentives to report higher financial accounting income; in contrast, the incentives shift under the tax reporting regime since companies want to report lower taxable income to reduce their tax liability.

Hypothesis 2 (H2). The negative association between external auditors' risk assessment and strong tax enforcement is greater when a country's book-tax conformity is high.

2.3.2.2 | *Auditor's exposure to complex tax issues*

Prior research suggests that complex tax issues often reside in the gray areas of tax laws, characterized by legal ambiguities that companies frequently exploit to structure sophisticated tax avoidance strategies (Balakrishnan et al., 2019; Desai & Dharmapala, 2006). For example, multinational companies exploit tax law differences across jurisdictions to obfuscate and separate income from underlying economic activities (Levin, 2013). As a result, companies with more complex tax matters often attract heightened scrutiny from tax authorities. For auditors working with these companies, this increased scrutiny may provide them with opportunities to observe the extensive verification efforts of the tax authority, thereby increasing their awareness of how the tax authority fulfills its monitoring role and curbs managerial opportunism. As auditors gain knowledge of tax enforcement through their exposure to complex tax issues, they begin to develop task-specific knowledge—a type of expertise shown to positively influence audit outcomes at both the individual and group levels (Bonner & Lewis, 1990; Goldman et al., 2022; Shepardson, 2019).

According to the concept of analogical reasoning (Shepardson, 2019), individuals integrate relevant experiences and information from one setting into other decision-making processes. Accordingly, exposure to complex tax issues can enhance how auditors incorporate the potential for tax enforcement into other tasks, such as assessing the client company's RMM. Supporting this argument, experimental evidence suggests that auditors' knowledge is transferable across tasks and industry contexts (Thibodeau, 2003). Therefore, we conjecture that auditors are more inclined to perceive potential monitoring benefits from tax enforcement when they have greater exposure to complex tax issues in their client base. We present our third hypothesis in the alternative form as follows:

Hypothesis 3 (H3). The negative association between external auditors' risk assessment and strong tax enforcement is greater when the auditor's exposure to complex tax issues is high.

3 | RESEARCH DESIGN AND SAMPLE

3.1 | Measuring tax enforcement

Consistent with prior research (Bauer et al., 2021; Guedhami & Pittman, 2008), we measure tax enforcement by estimating the likelihood of tax audits. While US studies have access to IRS audit rates, we rely on data from ISORA to capture the likelihood of tax audits across various countries. ISORA—with support from various international organizations, including the OECD—periodically collects tax administration data and publishes the survey results online. This survey asks tax administrations to self-report human resources, budget, revenue collections, management practices, and other related information (OECD, 2015, 2017a, 2019). ISORA provides data on the specific survey questions and responses, and the OECD uses these data to offer useful summaries and highlight trends in their *Tax Administration Comparative Information Series*.

To estimate the likelihood of tax audits, we focus on human capital resources dedicated to the audit and verification function. Our measure, *TAXENF*, is calculated as the total FTEs within the specific function listed as "audit, investigation, and other verification," scaled by inflation-adjusted GDP (in billions of dollars) in a given country and year.[6] Therefore, *TAXENF* allows us to compare the likelihood of tax audits across countries, holding each country's economy size constant. For ease of interpreting the regression coefficients, we divide the ratio by 10. Following De Simone et al. (2023), we measure tax enforcement using the country where the company's headquarters is located, since the company's primary business activities typically reside in that country.[7] Appendix S1 provides a timeline for variable construction as well as additional details regarding the ISORA data set used to measure tax enforcement.[8]

3.2 | Measuring auditor risk assessment

While risk assessment is a cornerstone of every audit engagement, this process has not been directly observable by outside parties. This lack of transparency occurs because the auditor's report has historically contained a boilerplate description and a pass/fail opinion. Under this format, users of the financial statements have very little insight into the audit, including risk assessment and subsequent fieldwork. Due to this data limitation, research to date often uses audit fees—or in some cases, audit hours or labor mix—to infer auditors' perceptions of client risk (Bae et al., 2021; Bell et al., 2008; Cho et al., 2017; Hogan & Wilkins, 2008; Niemi, 2002; O'Keefe et al., 1994). However, a recent change to the auditor's report presents an opportunity to learn more about auditors' perception of risk.

Over the last decade, regulators around the world have developed a renewed interest in providing more transparency into the audit process (Carcello, 2012; Institute of Chartered Accountants in England and Wales, 2017; Minutti-Meza, 2021). For example, in June 2013, the Financial Reporting Council (FRC) took the first step in expanding the auditor's report. This initiative required public disclosure of the most significant RMMs for companies with a premium listing on the London Stock Exchange. Similarly, for public companies listed in all other European Union countries with fiscal years ending on or after December 15, 2016, the IAASB issued ISA 701 to require communication of KAMs in the auditor's report.[9] Auditors must identify KAMs from matters communicated to those charged with governance and should consider areas with a heightened RMM and significant auditor judgment (IAASB, 2020c, para. 9). Once determined, KAM disclosures should describe the significant matter along with how the matter was addressed in the audit. Appendix S1 provides two examples of these KAM disclosures for companies in our sample.

In our analysis, we use the number of KAMs to capture the auditor's risk assessment (*NKAM*). The standards do not mandate a certain number of KAMs; rather, the number and type of KAMs should vary based on client-specific circumstances. Therefore, studies rely on variation in the number of KAMs to capture discretion in auditors' KAM reporting decisions (e.g., Rousseau & Zehms, 2024). The number of KAMs per audit opinion in our sample ranges

[6]In addition to FTEs focused on tax enforcement, FTEs are assigned to other roles that support the enforcement function. For example, based on 2018 data, Germany had more FTEs focused on audit, investigation, and other verification than the United States, while the United States had more FTEs engaged in supportive roles such as registration, taxpayer services, and returns and payment processing.

[7]While our sample represents companies listed on European exchanges, not all of these companies are headquartered in European countries as noted in Table 1 (Panel C). As a robustness test, we remove the observations for companies headquartered in non-European countries and continue to find consistent results.

[8]See Appendix S1 in the Supporting Information.

[9]The FRC further revised its standards in 2016 to fully converge with the IAASB standards to adopt the same definition of KAMs. The revised FRC rules became effective for all companies listed on the London Stock Exchange with fiscal year-ends on or after June 2017. In the United States, the PCAOB adopted a different version called CAMs, which took effect for audits of large accelerated filers with fiscal years ending on or after June 30, 2019, and for all other audits with fiscal years ending on or after December 15, 2020. Paragraph 11 of PCAOB Auditing Standard 3101 defines CAMs as "any matter arising from the audit of the financial statements that was communicated or required to be communicated to the audit committee and that (1) relates to accounts or disclosures that are material to the financial statements and (2) involved especially challenging, subjective, or complex auditor judgment" (PCAOB, 2017c).

from 1 to 11.[10] Existing research also reveals that audit partners have significant autonomy in reporting KAMs/CAMs based on the facts and circumstances of their clients (Dannemiller et al., 2025; Rousseau & Zehms, 2024).[11] Further supporting our measurement choice, studies show that the number of CAMs is associated with greater stock return volatility, more analyst dispersion, and lower financial reporting quality (Klevak et al., 2023; Li et al., 2025). Finally, Lynch et al. (2024) argue that KAMs can provide a confirmatory risk signal based on evidence that investors lower their valuation of tax avoidance and deferred tax assets when auditors issue a tax-specific KAM.

3.3 | Empirical model

To test our first prediction (H1), we estimate the following OLS regression:

$$\begin{aligned} NKAM_{it} = & \alpha_0 + \alpha_1 TAXENF_{jt} + \alpha_2 SIZE_{it} + \alpha_3 LEVERAGE_{it} + \alpha_4 \Delta SALE_{it} + \alpha_5 LOSS_{it} \\ & + \alpha_6 FOREIGN_{it} + \alpha_7 AUDITFEE_{it} + \alpha_8 BIG4_{it} + \alpha_9 VOL\_ETR_{it} + \alpha_{10} VOL\_OCF_{it} \\ & + \alpha_{11} VOL\_SPI_{it} + \alpha_{12} VOL\_PI_{it} + \alpha_{13} ETR_{it} + \alpha_{14} FRQ_{it} \\ & + \sum \alpha_k Industry/Country/Year + \varepsilon_{it}, \end{aligned} \tag{1}$$

where the dependent variable is the number of total KAMs (*NKAM*) disclosed in the auditor's report of company *i* in year *t*.[12] The variable of interest, *TAXENF*, is the tax enforcement measure of country *j* in year *t*. A negative coefficient on *TAXENF* would suggest that a heightened likelihood of tax audits is associated with fewer KAMs, supporting our prediction. In subsequent analysis, we disaggregate the dependent variable into tax KAMs and all other nontax KAMs. We follow Drake et al. (2024) and define tax KAMs (*NTAXKAM*) as those KAMs relating to deferred taxes, uncertain tax positions, or other taxes based on the taxonomy provided by Audit Analytics. *NOTHERKAM* represents all other KAM topics.

The control variables include a vector of company and auditor characteristics known to determine KAM disclosures. Specifically, prior research finds that operating complexity is associated with KAM disclosures (Burke et al., 2023). Therefore, we include the natural logarithm of total assets (*SIZE*), leverage (*LEVERAGE*), sales growth (*ΔSALE*), and negative earnings (*LOSS*) to control for operating complexity (Doyle et al., 2007; McGuire et al., 2012). Following De Simone et al. (2023), we use foreign exchange income/loss to approximate the existence of foreign operations (*FOREIGN*). Auditor type and effort also represent important factors associated with KAM disclosures (Rousseau & Zehms, 2024); thus, we create control variables that capture whether an auditor is a Big 4 firm (*BIG4*) as well as the natural logarithm of audit fees charged to the client (*AUDITFEE*).[13] Following Guenther et al. (2017), we construct several volatility measures, including the volatility of cash effective tax rates (*VOL_ETR*),

[10]The standards permit auditors to report no KAMs if appropriate; specifically, paragraph 16 of ISA 701 provides disclosure requirements for auditors who determine that there are no KAMs to communicate (IAASB, 2020c). However, Audit Analytics constructs the KAMs database by excluding company-year observations without any KAMs (refer to Section 3.4). Recent practitioner publications suggest that the frequency of no KAM observations is very small (e.g., less than 1%) (Audit Analytics, 2021). Therefore, consistent with this prior research, our sample does not include cases in which the auditor chooses to report no KAMs.

[11]Dannemiller et al. (2025) highlight that US auditors' biggest concern that might drive herding behavior in CAM reporting is their fear of PCAOB scrutiny. In contrast, auditors do not express concerns regarding pushback from management, the audit committee, or investors. Since the PCAOB inspection regime is not as salient for companies listed in Europe as it is in the United States, we expect greater KAM variation in our setting that improves the power of our tests. Moreover, Nylen et al. (2024) show that KAMs are incrementally informative to CAMs for the same company.

[12]We find consistent results (untabulated) using the Poisson Count model.

[13]We find consistent results (untabulated) if we exclude *AUDITFEE* from the model because it represents an alternative measure of auditor risk assessment in existing research.

volatility of cash flows (*VOL_OCF*), volatility of special items (*VOL_SPI*), and volatility of pre-tax income (*VOL_PI*) to control for different dimensions of company-specific risk that can increase tax and financial reporting complexity.[14] Finally, we control for tax avoidance (*ETR*) and financial reporting quality (*FRQ*) based on research indicating that tax enforcement is associated with both factors (De Simone et al., 2023; Hanlon et al., 2014; Hoopes et al., 2012).

Prior research documents that the institutional environment—specifically, the legal and investor protection environment—influences audit outcomes such as audit fees and the choice of Big 4 auditors (Choi et al., 2008; Choi & Wong, 2007; Francis & Wang, 2008). To control for these factors, we include country fixed effects since many of these institutional features are time-invariant during our sample period. In robustness tests described in Section 4, we include several additional time-varying country-level factors based on external parties noted in the auditing standards. The regression model also includes industry and year fixed effects to capture unobserved heterogeneity across industries and years.

Our cross-sectional predictions (H2 and H3) focus on whether the negative relation between auditors' risk assessment and country-level tax enforcement is more pronounced when there is greater book-tax conformity and when the auditor has more exposure to complex tax issues. We calculate book-tax conformity (*CONFORMITY*) based on the approach in Atwood et al. (2010). Consistent with their study, we require that a country has at least five observations with the necessary data to be included in our tests. To capture the auditor's exposure to complex tax issues (*AUDITOR_TAXEXP*), we follow Goldman et al. (2022) to identify three main areas of tax complexity that attract heightened scrutiny from the tax authority. First, R&D activities involve the calculation of R&D tax credits and the generation of intangible income, both of which are highly susceptible to tax avoidance. Second, multinational companies frequently shift income to jurisdictions with lower tax rates and establish obscure offshore entities to minimize tax. Third, M&A transactions present significant tax challenges around the determination of whether the transaction, or portions of it, is taxable. We separately use the presence of R&D spending, foreign operations, and goodwill to capture a company's tax complexity in these three areas. Following the approach of Goldman et al. (2022), we aggregate these factors and expect auditors to have greater exposure to complex tax issues when their client base exhibits more of these characteristics. The Appendix provides definitions for all variables used in our tests.[15] Each continuous variable is winsorized at the 1st and 99th percentiles prior to inclusion in the analysis.

3.4 | Sample selection

We obtain corporate financial information from Compustat Global and merge this data set with the Audit Analytics Europe Opinions and Fees data sets to obtain information about KAMs and auditor characteristics. Tax enforcement information comes from ISORA, as discussed previously. Finally, variables for other external parties that may also act as monitors are created and obtained from a variety of sources, including I/B/E/S, Reporters Without Borders, World Bank, and the OECD Trade Union data set.

Panel A of Table 1 summarizes the sample selection process. We begin our sample with 24,333 company-year observations that exist in the Audit Analytics Europe and Compustat Global data sets between the years 2014 and 2019. The sample period begins in 2014 and ends in 2019 due to the availability of the machine-readable tax enforcement data when our analysis commenced. We exclude 13,494 observations that are not listed on a European regulated exchange because Audit Analytics does not collect certain necessary data for these companies.

[14]We recognize that a related paper by Lynch et al. (2024, tab. 6) includes additional control variables in their tax KAM prediction model. While we do not expect these variables to correlate with country-level tax enforcement, we reperform our tests after including all controls in their model and find that our main results remain unchanged.

[15]Appendix S1 in the Supporting Information offers a timeline for the key variable measurements defined above.

TABLE 1 Sample selection and distribution.

Panel A: Sample selection

	Obs.
All company-year observations between 2014 and 2019 that are included in Compustat Global, Audit Analytics Europe Opinion, and Audit Analytics Europe Fee data sets	24,333
Less: Observations not in a European Union regulated market	(13,494)
Less: Observations not in an expanded audit report regime	(2,947)
Less: Observations not covered in the Audit Analytics KAMs data set	(2,842)
Less: Observations with missing country-level tax enforcement data	(136)
Less: Observations with missing data for control variables	(357)
Company-year observations in the final sample	4,557
Unique companies in the final sample	1,464

Panel B: Sample distribution by industry

Industry (two-digit SIC)	Frequency	Percent
Agriculture, forestry, and fishing (01–09)	24	0.53
Mining (10–14)	262	5.75
Construction (15–17)	206	4.43
Manufacturing (20–39)	2,158	47.36
Transportation and communication (40–47)	246	5.40
Utility (48–49)	297	6.52
Wholesale trade (50–51)	159	3.49
Retail trade (52–59)	241	5.29
Finance, insurance, and real estate (60–67)	4	0.09
Services (70–89)	908	20.00
Public administration (90–99)	52	1.14
Total	4,557	100.00

Panel C: Sample distribution by headquarter (HQ) country

HQ country	Frequency	Percent	*TAXENF*
United Kingdom	1,460	32.04	0.716
Germany	503	11.04	1.004
Sweden	490	10.75	0.597
France	426	9.35	0.450
The Netherlands	298	6.54	0.898
Finland	291	6.39	0.449
Norway	280	6.14	0.521
Denmark	188	4.13	0.557
Italy	173	3.80	0.684
Belgium	130	2.85	1.200
Austria	86	1.89	0.938
Ireland	57	1.25	0.459
Portugal	36	0.79	0.845
Luxembourg	26	0.57	1.037
Switzerland	23	0.50	0.038

TABLE 1 (Continued)

Panel C: Sample distribution by headquarter (HQ) country

HQ country	Frequency	Percent	*TAXENF*
Cyprus	16	0.35	1.142
Czech Republic	13	0.29	1.389
United States	13	0.29	0.116
Australia	7	0.15	0.385
Israel	7	0.15	0.610
Russian Federation	6	0.13	4.791
Spain	5	0.11	0.546
Hong Kong SAR	5	0.11	0.069
South Africa	5	0.11	0.664
Iceland	4	0.09	0.159
Singapore	4	0.09	0.109
Malaysia	3	0.06	1.204
Canada	2	0.04	0.640
	4,557	100.00	

Note: This table presents details on sample selection and distribution. Specifically, Panel A reports the sample selection process. Panel B reports the distribution of our sample by industry based on two-digit SIC codes. Finally, Panel C provides the mean values for tax enforcement (*TAXENF*) by the company's headquarter country for the companies in our sample that are listed on European exchanges.

Next, we drop 2,947 observations not included in the expanded auditor reporting regime. Another 2,842 observations are excluded because they do not exist in the Audit Analytics KAM data set.[16] Last, we remove 136 observations headquartered in countries outside the scope of the ISORA data set and remove 357 observations missing necessary control variables. These procedures result in a final sample of 4,557 company-year observations representing 1,464 unique companies.

Panels B and C of Table 1 show the sample distribution by industry and headquarter country, respectively. Focusing on Panel C, the three countries with the most observations in our sample are the United Kingdom, Germany, and Sweden. Of these three countries, Germany has the highest average tax enforcement ($TAXENF = 1.004$), surpassing both the United Kingdom (0.716) and Sweden (0.597). Economically, Germany has an average GDP of \$3,770 billion over the sample period, which translates into an average of 37,851 FTEs within the tax audit and verification function ($3,770 \times 1.004 \times 10$). In contrast, Sweden—with an average GDP of \$537 billion over the sample period—has an average of 3,206 FTEs in this function of the tax authority ($537 \times 0.597 \times 10$).

4 | RESULTS

4.1 | Univariate statistics

In Panel A of Table 2, we present descriptive statistics for the full sample. The mean (median) number of total KAMs reported is approximately 2.848 (3.000) across all companies in our

[16]We engaged in phone conversations with Audit Analytics personnel and manually inspected 100 company-year observations to understand this sample attrition. From our investigation, we learned the following: (1) the Audit Analytics KAMs data set only includes company-years with at least one KAM, excluding zero KAM observations; (2) Audit Analytics does not collect KAMs data for audit opinions published in a foreign (non-English) language; and (3) KAMs data may be missing from the database for other non-systemic reasons including human error or circumstances where the audit opinion is not machine-readable.

TABLE 2 Descriptive statistics.

Panel A: Descriptive statistics for full sample

	N	Mean	SD	P25	Median	P75
NKAM	4,557	2.848	1.339	2.000	3.000	4.000
NTAXKAM	4,557	0.254	0.450	0.000	0.000	0.000
NOTHERKAM	4,557	2.594	1.242	2.000	2.000	3.000
TAXENF	4,557	0.700	0.228	0.451	0.661	0.917
SIZE	4,557	7.278	2.090	5.843	7.233	8.696
LEVERAGE	4,557	0.746	1.337	0.158	0.492	0.969
ΔSALE	4,557	0.099	0.352	−0.013	0.053	0.134
LOSS	4,557	0.184	0.388	0.000	0.000	0.000
FOREIGN	4,557	0.685	0.465	0.000	1.000	1.000
AUDITFEE	4,557	13.488	1.474	12.416	13.367	14.454
BIG4	4,557	0.822	0.383	1.000	1.000	1.000
VOL_ETR	4,557	0.241	0.696	0.034	0.073	0.152
VOL_OCF	4,557	0.073	0.189	0.019	0.034	0.062
VOL_SPI	4,557	0.024	0.050	0.003	0.009	0.023
VOL_PI	4,557	0.077	0.193	0.017	0.033	0.068
ETR	4,557	−0.038	0.199	−0.046	−0.187	0.037
FRQ	4,557	0.458	0.335	0.444	0.111	0.778
Partitioning variables used in cross-sectional tests (raw values):						
CONFORMITY	4,544	0.416	0.164	0.308	0.400	0.514
AUDITOR_TAXEXP	4,557	129.244	96.377	45.000	113.500	207.000

Panel B: Comparison of observations across subsamples

	Strong tax enforcement subsample (*TAXENF_DUM* = 1)			Weak tax enforcement subsample (*TAXENF_DUM* = 0)			
	N	Mean	SD	*N*	Mean	SD	Difference
NKAM	2,186	2.930	1.314	2,371	2.772	1.357	0.158***
NTAXKAM	2,186	0.285	0.460	2,371	0.226	0.438	0.059***
NOTHERKAM	2,186	2.638	1.219	2,371	2.553	1.262	0.085**
SIZE	2,186	7.136	2.052	2,371	7.408	2.116	−0.272***
LEVERAGE	2,186	0.766	1.312	2,371	0.729	1.360	0.037
ΔSALE	2,186	0.096	0.337	2,371	0.103	0.365	−0.007
LOSS	2,186	0.174	0.379	2,371	0.194	0.396	−0.020*
FOREIGN	2,186	0.652	0.476	2,371	0.715	0.452	−0.063***
AUDITFEE	2,186	13.455	1.402	2,371	13.519	1.537	−0.064
BIG4	2,186	0.891	0.312	2,371	0.757	0.429	0.134***
VOL_ETR	2,186	0.247	0.710	2,371	0.235	0.684	0.012
VOL_OCF	2,186	0.071	0.195	2,371	0.074	0.184	−0.003
VOL_SPI	2,186	0.023	0.046	2,371	0.025	0.053	−0.002
VOL_PI	2,186	0.069	0.170	2,371	0.084	0.212	−0.015***
ETR	2,186	−0.037	0.197	2,371	−0.038	0.201	0.001
FRQ	2,186	0.478	0.330	2,371	0.440	0.339	0.038***

Note: This table presents the descriptive statistics for regression model variables in the full sample and for the cross-sectional tests (Panel A) as well as separately by the strength of country-level tax enforcement (Panel B). In Panel B, the stronger tax enforcement subsample is defined as those with values above the sample median using *TAXENF_DUM*. All variables are defined in the Appendix.

sample, which is consistent with other studies in the United Kingdom (Rousseau & Zehms, 2024) but slightly higher relative to studies in other countries such as the United States and Hong Kong Special Administrative Region (SAR) (Burke et al., 2023; Liao et al., 2024). When we separate KAMs into tax (nontax) topics, the mean value for *NTAXKAM* (*NOTHERKAM*) is 0.254 (2.594) with a maximum value of 2.000 (10.000). Our test variable of interest (*TAXENF*) has a mean value of 0.700.

In Panel B of Table 2, we partition the sample by strong versus weak tax enforcement using the median value for tax enforcement in our sample (*TAXENF_DUM*). By comparing the mean values of key regression variables across both groups, we find that companies headquartered in countries with strong tax enforcement, on average, report slightly more KAMs (*NKAM*), are smaller (*SIZE*), are less likely to report a loss (*LOSS*), have fewer foreign operations (*FOREIGN*), are more likely to hire a Big 4 auditor (*BIG4*), exhibit less volatility in pre-tax income (*VOL_PI*), and exhibit higher financial reporting quality (*FRQ*). These differences highlight the importance of including these characteristics as control variables in our multivariate tests.

4.2 | Tests of the first prediction (H1)

4.2.1 | Primary model and alternative specifications

Table 3 reports the main results for our first prediction (H1) using Equation (1). In Column 1, we find that as the likelihood of tax audits increases, the number of total KAMs decreases (coeff. = −1.028, *p*-value <0.01). The control variables in the model also generally behave consistently with prior research (Burke et al., 2023; Lynch et al., 2024). We find that company size (*SIZE*), the ratio of debt to assets (*LEVERAGE*), the presence of a loss (*LOSS*), audit fees (*AUDITFEE*), and volatility in reporting of special items (*VOL_SPI*) are associated with more KAMs disclosed in the auditor's report. Moreover, higher values of *ETR* and *FRQ* (i.e., less tax avoidance and better financial reporting quality) are associated with fewer KAMs disclosed in the auditor's report. After controlling for these factors, the baseline model supports our prediction that auditors perceive less RMM in the presence of strong country-level tax enforcement. Overall, the identification and disclosure of fewer KAMs for companies in countries with a heightened likelihood of tax audits suggest that auditors perceive strong tax enforcement as an effective monitor when assessing the company's risk.

In Columns 2–7, we demonstrate that our main results are robust to a series of alternative specifications. To mitigate concerns that our results are driven by time-invariant, company-specific characteristics, we show that our results continue to hold in Column 2 after including company fixed effects (coeff. = −0.623, *p*-value <0.01). The sample is slightly smaller in this specification since we exclude singleton observations (Breuer & deHaan, 2024). Column 3 further demonstrates the robustness of the main finding when we use a first-order change specification (coeff. = −1.183, *p*-value <0.01). The change specification alleviates the concern that country-level or company-level trends might drive our results and further addresses the first-order autocorrelation issue. As an alternative (untabulated) form of this change test, we identify the largest increase in country-level tax enforcement (30.3% in Norway from 2018 to 2019) and the largest decrease in country-level tax enforcement (−39.9% in the United Kingdom from 2017 to 2018) based on the countries with a sufficient number of observations (i.e., at least 100). We construct a separate indicator variable to capture the large increase (decrease) and use this variable to replace *TAXENF* in the change specification. Consistent with our prediction in H1, the results reveal that a large increase in *TAXENF* is associated with a significant decrease in KAMs (coeff. = −0.160, *p*-value <0.10), whereas a large decrease in *TAXENF* is associated with a significant increase in KAMs (coeff. = 0.332, *p*-value <0.01).

As is common in cross-country studies, our setting raises the concern that unobservable time-varying factors may correlate with country-level tax enforcement and contribute to auditors' risk

TABLE 3 Tax enforcement and KAMs.

Dependent variable:	*NKAM*	*NKAM*	*NKAM*	*TAXENF*	*NKAM*	*NKAM*	*NKAM*	*NKAM*
	Baseline	Company FE	Changes	Instrumental variable (2SLS)		Entropy balancing	First-time KAM adopters	Excluding RMM obs.
				First stage	Second stage			
	(1)	(2)	(3)	(4a)	(4b)	(5)	(6)	(7)
TAXENF	−1.028***	−0.623***	−1.183***		−1.258*		−2.873***	−1.349***
	(0.193)	(0.177)	(0.176)		(0.715)		(0.505)	(0.192)
TAXENF_DUM						−0.409***		
						(0.057)		
HIGH_MILITARY				−0.094***				
				(0.004)				
SIZE	0.099***	0.361***	−0.045	0.001	0.099***	0.089***	0.038	0.090***
	(0.026)	(0.099)	(0.115)	(0.001)	(0.026)	(0.025)	(0.048)	(0.026)
LEVERAGE	0.037**	0.057***	−0.016	0.001	0.037**	0.054***	0.009	0.019
	(0.017)	(0.020)	(0.032)	(0.001)	(0.017)	(0.020)	(0.038)	(0.018)
ΔSALE	0.021	0.050	0.033	−0.003	0.020	0.000	0.046	0.075
	(0.050)	(0.049)	(0.047)	(0.003)	(0.051)	(0.055)	(0.142)	(0.052)
LOSS	0.323***	0.253***	0.132**	−0.002	0.322***	0.319***	0.104	0.306***
	(0.061)	(0.065)	(0.063)	(0.003)	(0.061)	(0.063)	(0.127)	(0.061)
FOREIGN	−0.042	0.113	0.119	−0.000	−0.042	−0.081	−0.036	−0.052
	(0.061)	(0.094)	(0.085)	(0.002)	(0.061)	(0.063)	(0.103)	(0.063)
AUDITFEE	0.276***	0.340***	0.371***	−0.004***	0.274***	0.297***	0.308***	0.256***
	(0.035)	(0.086)	(0.086)	(0.001)	(0.036)	(0.036)	(0.063)	(0.035)
BIG4	−0.130	−0.399***	−0.341**	0.008***	−0.128	−0.115	−0.041	0.018
	(0.083)	(0.152)	(0.172)	(0.003)	(0.084)	(0.090)	(0.173)	(0.083)
VOL_ETR	0.043	−0.023	0.010	−0.002	0.043	0.066**	0.117	0.034
	(0.033)	(0.041)	(0.063)	(0.001)	(0.033)	(0.033)	(0.072)	(0.031)

TABLE 3 (Continued)

Dependent variable:	*NKAM*	*NKAM*	*NKAM*	*TAXENF*	*NKAM*	*NKAM*	*NKAM*	*NKAM*
	Baseline	Company FE	Changes	Instrumental variable (2SLS)		Entropy balancing	First-time KAM adopters	Excluding RMM obs.
				First stage	Second stage			
	(1)	(2)	(3)	(4a)	(4b)	(5)	(6)	(7)
VOL_OCF	0.050	−0.123	−0.158	−0.003	0.050	0.146	0.792*	0.013
	(0.201)	(0.239)	(0.334)	(0.006)	(0.201)	(0.228)	(0.406)	(0.205)
VOL_SPI	1.382***	0.952	0.475	0.005	1.385***	0.922*	1.566	1.160**
	(0.478)	(0.766)	(0.877)	(0.020)	(0.479)	(0.547)	(1.001)	(0.508)
VOL_PI	−0.084	0.032	0.423	0.001	−0.085	−0.249	−0.640*	0.054
	(0.185)	(0.232)	(0.399)	(0.009)	(0.185)	(0.228)	(0.331)	(0.178)
ETR	−0.227**	−0.077	0.016	−0.010*	−0.230**	−0.235**	−0.350	−0.241**
	(0.103)	(0.089)	(0.105)	(0.005)	(0.103)	(0.111)	(0.228)	(0.105)
FRQ	−0.105*	−0.071	−0.022	0.002	−0.105*	−0.070	0.046	−0.105*
	(0.059)	(0.051)	(0.049)	(0.003)	(0.059)	(0.061)	(0.140)	(0.059)
Intercept	−0.540	−3.774***	3.850***	0.491***		−1.203**	2.422***	−0.557
	(0.643)	(1.189)	(0.092)	(0.015)		(0.572)	(0.872)	(0.659)
Country FE	Yes	No	No	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	No	No	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
Company FE	No	Yes	No	No	No	No	No	No
N	4,557	4,376	3,747	4,557	4,557	4,557	651	3,772
R^2	0.409	0.770	0.097	0.913	0.195	0.420	0.419	0.383

Note: This table presents the regression results to examine the association between tax enforcement (*TAXENF*) and the disclosure of KAMs in the auditor's report. The dependent variable across all specifications, with the exception of the first-stage analysis in Column 4a, is the number of KAMs (*NKAM*). Column 1 provides the results with our main specification, whereas Columns 2–7 provide alternative specifications for robustness. Specifically, the robustness tests include the following: Column 2 includes company and year fixed effects; Column 3 provides a changes specification; Columns 4a and 4b provide regression results using a 2SLS instrumental variables approach in which the instrument in the first-stage regression is *HIGH_MILITARY*; Column 5 includes the results after entropy balancing based on *TAXENF_DUM*; Column 6 provides the results when only including the first-time adopters of KAM disclosures under the official ISA 701 definition; and Column 7 reports the results after excluding UK observations prior to 2017 under the RMM standard. All variables are defined in the Appendix. Robust standard errors clustered by company are included in parentheses.
***, **, and * represent significance at 0.01, 0.05, and 0.1 levels, respectively, based on two-tailed tests.

assessment. To address this concern, we instrument country-level tax enforcement with country-level military spending in a two-stage least squares (2SLS) approach. Conventional wisdom emphasizes that government spending faces a trade-off between military expenses ("guns") and social activities ("butter"). Political candidates often use "guns" versus "butter" to differentiate their ideologies. Existing research provides some evidence that increases in military spending come at the cost of spending on other governmental activities (Russett, 1969; R. P. Smith, 1977). Based on military spending data from the World Bank, we expect a country's military expenditures to exhibit a negative association with investment in the tax authority's human resources. In contrast, it is difficult to establish a direct link between a country's military spending and auditors' assessment of misstatement risk at individual companies. We measure military spending using an indicator variable equal to one if the ratio of military spending to GDP for country i in year t is in the top quartile of the sample (*HIGH_MILITARY*), and zero otherwise.

We present our first-stage regression results in Column 4a of Table 3, which support these arguments and demonstrate a negative and significant association between high levels of military spending and tax enforcement (coeff. $= -0.094$, p-value <0.01). Further, the Stock-Yogo test suggests that military spending is strongly correlated with tax enforcement, which establishes the relevance and strength of our instrument (F-statistic $= 46.27$).[17] The second-stage regression is reported in Column 4b and indicates that the negative and significant association between *TAXENF* and *NKAM* continues to hold when we instrument *TAXENF* with *HIGH_MILITARY* (coeff. $= -1.258$; p-value <0.10). While these tests cannot fully mitigate the endogeneity concern, they help to support the specific link between tax enforcement and auditors' risk assessment through KAM reporting decisions.

As previously shown in Panel B of Table 2, companies in strong versus weak tax enforcement countries exhibit different characteristics. To address this covariate imbalance, we report our results using entropy balancing techniques in Column 5. Whereas propensity score matching assigns a weight of either zero or one to each control observation, entropy balancing estimates continuous weights for observations in the control sample (i.e., weak tax enforcement) to achieve covariate balance, ensuring that we retain all observations in the full sample. We require covariate balance on the first, second, and third moments of the distributions of all covariates across the two subsamples based on observations that are above and below the median value for *TAXENF* (i.e., using the indicator variable *TAXENF_DUM*). In doing so, we find that the weight ratio is 0.14 and the maximum weight is 3.60 in the entropy balanced sample. Thus, we confirm that entropy balancing did not assign excessively high weights to a large number of control observations (McMullin & Schonberger, 2020, 2022). We continue to find that our core evidence holds (coeff. $= -0.409$, p-value <0.01) using this entropy balanced sample.

Since the United Kingdom required auditors to report RMMs prior to adopting KAMs under ISA 701, our sample includes company-year observations reporting under two different regimes. As noted in Minutti-Meza (2021), "RMMs in the 2013 rules and KAMs in the 2016 rules are largely similar" such that we expect expanded reports to be relatively stable following the first year of implementation (p. 553). However, to test the robustness of our results to different reporting regimes, we reperform our tests using a sample of auditors reporting KAMs for the first time under the official ISA 701 definition adopted by IAASB. The sample in Column 6 consists of companies with fiscal year-ends in 2017 for the United Kingdom and the Netherlands and fiscal year-ends in 2016 for all other countries. We focus on first-time KAMs since we expect the initial selection and reporting of KAMs requires the most professional judgment. Given the limited time periods in this specification, we do not include year fixed effects in this test. Related, in Column 7, we remove observations from

[17] The Stock-Yogo test looks at the first-stage F-test to see if the instrument is weak. If the F-statistic is above 8.96 (one instrument), the instrument is considered strong (Larcker & Rusticus, 2010). Meanwhile, we cannot directly test the validity of the instrument (i.e., whether it satisfies the exclusion restriction) because of the equal number of instruments and endogenous variables. Further, we note that the regression results in Table 3 do not include an intercept because we use the "partial" option with the ivreg2 command in Stata.

the United Kingdom before 2017 to exclude audit reports with RMMs from the sample. Consistent with our main results, we find tax enforcement is negatively associated with the total number of KAMs using this sample of first-time KAM adopters (coeff. = −2.873, p-value <0.01) and the sample without RMMs from the United Kingdom (coeff. = −1.349, p-value <0.01).[18]

4.2.2 | Additional test based on the type of KAM

To gain a deeper understanding of whether tax enforcement is salient to auditors' risk assessment of tax accounts as well as other financial statement accounts, we estimate Equation (1) using tax and nontax KAMs in Table 4. Column 1 shows a negative and significant association between the likelihood of tax audits and tax-specific KAMs ($NTAXKAM$) (coeff. = −0.134, p-value <0.05). This result has intuitive appeal as it suggests tax enforcement reduces auditors' perceptions of tax-specific risk. Turning to our analysis of nontax risk, the results in Column 2 suggest that the likelihood of tax audits is also negatively associated with auditors' reporting of other types of KAMs ($NOTHERKAM$) (coeff. = −0.940, p-value <0.01).[19] Following our discussion in Section 2, tax authorities extensively scrutinize business records from the company's internal information system and various third parties (e.g., banks, online payment systems, and suppliers) to audit tax return information. These enforcement actions help verify accounting records and strengthen the monitoring role of tax authorities, which influences how auditors perceive the risk of other accounts in financial statements. This evidence on the negative association between strong tax enforcement and auditors' assessment of nontax risk also extends and complements the findings in Chow et al. (2022), who demonstrate that tax enforcement intensity is associated with fewer tax-related misstatements as well as other accounting misstatements.[20]

4.3 | Cross-sectional tests (H2 and H3)

Following the cross-sectional predictions in Section 2, we next examine whether the effect of tax enforcement on auditors' assessment of the RMM is stronger when there is greater book-tax conformity in the country (Table 5) and when the auditor has greater exposure to complex tax issues (Table 6). We measure each cross-sectional partitioning variable based on the average value in the prior and current year. In all tests, we compare observations in the top quartile to observations in the bottom quartile of each partitioning variable to isolate high versus low levels of that characteristic. We use seemingly unrelated regressions to assess the significance of coefficient differences (chi-squared) between high and low subsamples.[21]

[18]Given the variation in $TAXENF$ by country as noted in Panel C of Table 1, we also sequentially drop each country one by one from the sample and find (untabulated) that our primary results hold in all cases (p-value <0.01).

[19]The correlation between $NTAXKAM$ and $NOTHERKAM$ is 0.365, indicating that the two types of KAMs are related but not to a high degree. In further (untabulated) tests, we include $NOTHERKAM$ ($NTAXKAM$) as an additional control variable in the model when the dependent variable is $NTAXKAM$ ($NOTHERKAM$). We continue to see a negative coefficient on $TAXENF$ with a similar coefficient magnitude in these specifications (p-value <0.01). Additionally, we find a negative and significant coefficient on $NOTHERKAM$ ($NTAXKAM$) in these separate regressions. This evidence alleviates the concern that the likelihood of tax audits influences tax (nontax) KAMs through nontax (tax) KAMs.

[20]Appendix S1 in the Supporting Information provides robustness tests using alternative measures for tax enforcement (based on budget resources) and KAMs (based on uncertain language and the length of the KAM description).

[21]We examine the sensitivity of these results using several iterative specifications that modify the cutoff for the subsample partition (i.e., terciles instead of quartiles) as well as the measurement timing of the cross-sectional variables (i.e., based only on prior year or current year data instead of the average of the two periods). We find that the conclusions reported in Tables 5 and 6 remain consistent across most alternative specifications. Nevertheless, we notice a lack of significance for $TAXENF$ when measuring $CONFORMITY$ based only on prior year data and using quartiles for the partitions. Similarly, using the same time horizon and partitioning for $AUDITOR\_TAXEXP$, we find that the coefficient for $TAXENF$ remains negative and significant in the subsample with high auditor exposure, but the cross-equation comparison is somewhat weaker ($\chi^2 = 2.22$). This evidence suggests that it may be important to incorporate current year information for auditors to observe and update their judgments.

TABLE 4 Tax enforcement and the types of KAMs.

Dependent variable:	*NTAXKAM* (1)	*NOTHERKAM* (2)
TAXENF	−0.134**	−0.940***
	(0.065)	(0.190)
SIZE	−0.008	0.106***
	(0.010)	(0.025)
LEVERAGE	0.012	0.026
	(0.008)	(0.018)
ΔSALE	−0.048***	0.067
	(0.016)	(0.050)
LOSS	0.026	0.301***
	(0.023)	(0.059)
FOREIGN	0.009	−0.042
	(0.025)	(0.060)
AUDITFEE	0.105***	0.166***
	(0.014)	(0.035)
BIG4	0.044	−0.174**
	(0.031)	(0.085)
VOL_ETR	0.033**	0.011
	(0.014)	(0.030)
VOL_OCF	−0.094	0.076
	(0.077)	(0.226)
VOL_SPI	0.235	1.075**
	(0.203)	(0.498)
VOL_PI	0.071	−0.076
	(0.078)	(0.188)
ETR	−0.068	−0.165*
	(0.043)	(0.098)
FRQ	−0.058***	−0.048
	(0.022)	(0.056)
Intercept	−0.516**	0.112
	(0.256)	(0.671)
Country, industry, year FE	Yes	Yes
N	4,557	4,557
R^2	0.224	0.348

Note: This table presents the regression results to examine the association between tax enforcement and tax-related KAMs (*NTAXKAM*) versus all other nontax KAMs (*NOTHERKAM*). All variables are defined in the Appendix. Robust standard errors clustered by company are included in parentheses.
***, **, and * represent significance at the 0.01, 0.05, and 0.1 levels, respectively, based on two-tailed tests.

Table 5 reports the subsample results based on countries with high versus low book-tax conformity (*CONFORMITY*). In Columns 1 and 2, using *NKAM* as the dependent variable, the regression results reveal that the negative association between the likelihood of tax audits and KAMs is concentrated in the subsample with high book-tax conformity, as predicted in H2. The chi-squared test also indicates that the coefficient for *TAXENF* is statistically different

TABLE 5 Cross-sectional test: Book-tax conformity.

Dependent variable:	***NKAM***		***NTAXKAM***		***NOTHERKAM***	
	High book-tax conformity (1)	**Low book-tax conformity (2)**	**High book-tax conformity (3)**	**Low book-tax conformity (4)**	**High book-tax conformity (5)**	**Low book-tax conformity (6)**
TAXENF	−2.153***	0.102	−0.294**	0.136	−1.911***	0.269
	(0.346)	(0.882)	(0.120)	(0.436)	(0.346)	(1.028)
Test of diff. (χ^2) $TAXENF^{High} = TAXENF^{Low}$	6.12**		0.98		4.35**	
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Country, industry, year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1,238	1,185	1,238	1,185	1,238	1,185
R^2	0.414	0.368	0.311	0.202	0.351	0.356

Note: This table presents the cross-sectional tests related to book-tax conformity (*CONFORMITY*) following the approach in Atwood et al. (2010). Specifically, we create separate subsamples for high (top quartile) and low (bottom quartile) levels of *CONFORMITY* and compare the coefficient on *TAXENF* across these two subsamples. Observations in the middle two quartiles of the main sample are removed from this test. All variables are defined in the Appendix. Robust standard errors clustered by company are included in parentheses.
***, **, and * represent significance at 0.01, 0.05, and 0.1 levels, respectively, based on two-tailed tests.

TABLE 6 Cross-sectional test: Auditor exposure to complex tax issues.

Dependent variable:	***NKAM***		***NTAXKAM***		***NOTHERKAM***	
	High auditor exposure (1)	**Low auditor exposure (2)**	**High auditor exposure (3)**	**Low auditor exposure (4)**	**High auditor exposure (5)**	**Low auditor exposure (6)**
TAXENF	−1.149***	−0.227	−0.119	−0.169	−1.131***	−0.008
	(0.418)	(0.406)	(0.124)	(0.158)	(0.406)	(0.398)
Test of diff. (χ^2) $TAXENF^{High} = TAXENF^{Low}$	2.72*		0.07		4.22**	
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Country, industry, year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1,140	1,142	1,140	1,142	1,140	1,142
R^2	0.482	0.478	0.372	0.312	0.398	0.430

Note: This table presents the cross-sectional tests related to the auditor's exposure to complex tax issues in its client base (*AUDITOR_TAXEXP*) following the approach in Goldman et al. (2022). Specifically, we create separate subsamples for high (top quartile) and low (bottom quartile) levels of *AUDITOR_TAXEXP* and compare the coefficient on *TAXENF* across these two subsamples. Observations in the middle two quartiles of the main sample are removed from this test. All variables are defined in the Appendix. For this table only, we include three additional control variables: *R&D* is the ratio of R&D expense to total assets and is set to zero if missing; *GOODWILL* is the ratio of goodwill to total assets and is set to zero if missing; and *OFFICE_SIZE* is the natural logarithm of the number of clients in the audit firm-office. Robust standard errors clustered by company are included in parentheses.
***, **, and * represent significance at 0.01, 0.05, and 0.1 levels, respectively, based on two-tailed tests.

across subsamples (p-value <0.05). When we separately analyze *NTAXKAM* versus *NOTHERKAM*, we continue to find that the negative association is concentrated in the high book-tax conformity subsamples, as reported in Columns 3 and 5. However, the coefficients are only statistically different across subsamples when *NOTHERKAM* is the dependent variable (p-value <0.05). Thus, when book-tax conformity is greater, auditors are more likely to consider the potential monitoring benefits of tax enforcement when evaluating risk in nontax accounts, due to the reporting consistency between the tax and financial reporting systems. In contrast, book-tax conformity appears less likely to influence the auditor's perception when evaluating the RMM in tax accounts.

Table 6 presents the subsample results based on auditors exhibiting high versus low exposure to complex tax issues in their client base (*AUDITOR_TAXEXP*). In this model, we include additional control variables to capture the individual client's likelihood of attracting tax regulatory involvement using *R&D* and *GOODWILL*—along with *FOREIGN*, which is already included in the primary specification in Equation (1). *R&D* (*GOODWILL*) is measured based on the reported R&D expense (goodwill asset) scaled by the total assets of company i in year t and is set to zero if missing. We also include a control for *OFFICE_SIZE* to ensure that our measure of auditor tax exposure is distinct from the size of the audit office. *OFFICE_SIZE* is measured as the natural logarithm of the number of clients in the audit firm-office in year t. The mean values (untabulated) of *R&D*, *GOODWILL*, and *OFFICE_SIZE* are 0.031, 0.157, and 3.663, respectively.

In Columns 1 and 2 of Table 6, where the dependent variable is *NKAM*, we find that the negative association between the likelihood of tax audits and KAMs is only significant in the high tax exposure subsample, as predicted in H3. Moreover, the chi-squared test shows that the coefficient for *TAXENF* is statistically different in the high exposure subsample relative to the low exposure subsample (p-value <0.10). These results suggest that auditors are more inclined to perceive the tax authority's monitoring role when they have been exposed to complex tax issues in their client base. When we separately use *NTAXKAM* and *NOTHERKAM* as the dependent variables, we find that the negative association is concentrated in the high exposure subsample only for *NOTHERKAM* as shown in Column 5. The coefficients in this specific test are also statistically different across subsamples (p-value <0.05), whereas the difference is insignificant when *NTAXKAM* is the dependent variable. This pattern of evidence suggests that tax enforcement may not incrementally inform auditors' perception of tax-specific misstatement risk when they already possess tax task-specific knowledge. Instead, their exposure to complex tax issues may familiarize them with the verification effort of the tax authority, such that they incorporate this information into their perceptions of misstatement risk in other financial statement accounts. However, we cannot rule out that this result is instead driven by a lack of power, given the low frequency of tax KAMs.[22] Collectively, these cross-sectional tests further support our primary results by showing unique channels through which the potential for tax enforcement could influence auditors' perception of misstatement risk, especially for nontax accounts.

4.4 | Additional robustness test considering other external parties

Besides the tax authority, the auditing standards reference others who may review and analyze the company's financial performance. Specifically, paragraph A79 of ISA 315 lists the following external parties: analysts, news and media, other regulators, trade unions, and providers of finance (IAASB, 2020b). Auditors may derive information from these other parties in their risk assessment, and the ability of these parties to act as alternative monitors could be correlated with our tax enforcement variable. Therefore, it is important to assess whether controlling for these other parties affects our main findings. To construct country-year measures for each

[22]The results (untabulated) are similar if we use alternative measures for the control variables related to R&D, goodwill, and audit office size. Specifically, we create indicator variables for the presence of R&D and goodwill instead of the continuous versions of these control variables noted previously. We also redefine office size based on total fees instead of total clients. In this robustness test, the coefficients on *TAXENF* and the test of coefficient difference across subsamples exhibit consistent magnitudes and significance levels to those reported in Table 6.

external party, we obtain data from a variety of sources noted in the Appendix, including I/B/E/S, World Bank, OECD, and Reporters Without Borders (Adams et al., 2024; Chyz et al., 2013; Qi et al., 2010; Shroff et al., 2014). We capture the influence of analysts on the information environment using the median analyst following (*ANALYST*). Next, we use freedom of the press to measure media monitoring (*FREE_PRESS*). *REG_QUALITY* captures the perceived strength of outside regulations in permitting and promoting private sector development. We capture the fourth external party using *TRADE_UNION* as a measure of variation in union influence. Finally, *EASE_OF_CREDIT* and *VENTURE_CAP* represent providers of finance since these measures reflect financing obtained through both lenders and venture capitalists. We do not offer a directional prediction on how other external parties influence auditors' risk perception.

Table 7 presents the regression results after including these variables as additional controls. Given the need for data from multiple sources, our sample declines to 4,096 company-year observations. When including these additional time-varying factors, the likelihood of tax audits continues to exhibit a negative and statistically significant association with the number of KAMs in the auditor's report when the dependent variable is *NKAM* (coeff. = −0.923, p-value <0.01), and *NOTHERKAM* (coeff. = −0.835, p-value <0.01). However, the results using *NTAXKAM* as the dependent variable just narrowly miss the traditional significance level (coeff. = −0.115, p-value = 0.11). Notably, the largest correlation between any of these other external parties and *TAXENF* is 0.455 (*TRADE_UNION*), and the largest variance inflation

TABLE 7 Controlling for other external parties noted in the auditing standards.

Dependent variable:	*NKAM* (1)	*NTAXKAM* (2)	*NOTHERKAM* (3)
TAXENF	−0.923***	−0.115	−0.835***
	(0.207)	(0.072)	(0.205)
ANALYST	0.103	−0.036	0.138*
	(0.071)	(0.024)	(0.071)
FREE_PRESS	0.003	0.000	0.001
	(0.008)	(0.003)	(0.007)
REG_QUALITY	−0.044***	−0.009	−0.033**
	(0.017)	(0.006)	(0.017)
TRADE_UNION	0.070***	0.009	0.061***
	(0.021)	(0.009)	(0.020)
EASE_OF_CREDIT	0.002	0.003**	−0.001
	(0.003)	(0.001)	(0.003)
VENTURE_CAP	0.007	0.001	0.006
	(0.006)	(0.002)	(0.006)
Control variables	Yes	Yes	Yes
Country, industry, year FE	Yes	Yes	Yes
N	4,096	4,096	4,096
R^2	0.421	0.233	0.358

Note: This table presents the regression results to examine the association between tax enforcement and the disclosure of KAMs, after controlling for other external parties cited in the auditing standards that may act as additional monitors. The other external parties in a specific country that we include are analysts (*ANALYST*), news and other media (*FREE_PRESS*), other regulators (*REG_QUALITY*), trade unions (*TRADE_UNION*), and providers of finance (*EASE_OF_CREDIT* and *VENTURE_CAP*). All variables are defined in the Appendix. Robust standard errors clustered by company are included in parentheses.

***, **, and * represent significance at 0.01, 0.05, and 0.1 levels, respectively, based on two-tailed tests.

factor (VIF) is 3.60.[23] Consequently, these alternative measures appear to capture constructs that differ from tax enforcement and do not influence our main findings when included in the model.

5 | CONCLUSION

International accounting standard setters emphasize the importance of using external factors to inform auditors' risk assessment. We focus on the tax authority's enforcement efforts as evidence of strong external monitoring that auditors may consider when assessing a client's RMM. Though the risk assessment process is traditionally private, the adoption of KAMs by companies listed in European countries offers a valuable opportunity to assess whether and how auditors consider the role of tax enforcement in their assessment of misstatement risk. Concurrent with this change to the auditor's report, the OECD has been working with countries to strengthen tax enforcement to combat tax avoidance. This setting provides a strong testing ground to examine whether the presence of strong tax enforcement influences auditors' risk assessment.

We find that auditors perceive less heightened RMMs by issuing fewer total KAMs when country-level tax enforcement is strong. Our results are robust to a series of additional tests, such as controlling for other external factors referenced in the auditing standards and employing a changes specification. Further, this association exists for both tax and nontax KAMs, suggesting the effect of tax enforcement on informing auditors' risk assessment extends beyond the tax account. The cross-sectional tests based on book-tax conformity and the auditor's exposure to complex tax issues provide additional context to these results.

Our analysis is subject to several caveats. First, although we provide robust analysis on the association between tax enforcement and auditors' risk assessment, it is inherently difficult to infer causality from our tests. Thus, we encourage readers to interpret our results with this limitation in mind. Second, since we cannot directly observe the strength of tax enforcement, we adopt a measure based on FTEs in the audit and verification function of the country's tax authority. This measure is intuitive as it captures the perceived likelihood of tax audits based on employees dedicated to performing this work. However, we recognize that this measure may not fully capture the underlying construct. Finally, while our setting covers a broad range of companies listed on European exchanges, we cannot know with certainty that our evidence will generalize to other contexts such as the United States with a stricter regulatory and legal regime and a different CAM definition.

Overall, our research contributes to the literature on the externalities of tax enforcement for outside stakeholders, sheds light on the auditors' risk assessment process, and adds to our understanding of the KAM reporting regime. We provide timely insights for regulators worldwide to assess whether factors in the external environment inform auditors' identification and disclosure of KAMs. Our study also extends the literature on how cross-country institutional differences influence auditor behaviors (e.g., Francis & Wang, 2008). We believe this evidence is important as international regulators increasingly coordinate their policy actions.

ACKNOWLEDGMENTS

We thank our editor, Dechun Wang, and two anonymous reviewers for their insightful guidance. We also appreciate the constructive comments on earlier versions of our paper from Matt Beck, Justin Blann, Robert Davidson, Marcus Doxey, Matt Erickson, Nathan Goldman (discussant), Michelle Harding, Jeff Pittman, Ulrike Thuerheimer, workshop participants at

[23]The typical threshold used to detect "harmful multicollinearity" is a VIF greater than 10 (Kennedy, 2008, p. 199). Thus, the VIFs for this analysis are below the threshold and do not indicate multicollinearity threats.

Virginia Tech, and conference participants at the 2022 American Taxation Association Midyear Meeting, the 2022 International Symposium on Audit Research, and the 2022 American Accounting Association Annual Meeting.

DATA AVAILABILITY STATEMENT

Data are publicly available from the sources identified in the paper.

ORCID

Jessica R. Filosa https://orcid.org/0000-0002-4353-0584
Jingjing (Jing) Huang https://orcid.org/0000-0003-2220-7494
Lijun (Gillian) Lei https://orcid.org/0000-0001-7742-5750
Sarah E. Stein https://orcid.org/0000-0002-5650-3147

REFERENCES

Adams, J., Demers, E., & Klassen, K. (2024). *Tax aggressive behavior and voluntary tax disclosure: Evidence from corporate sustainability reporting*. Working paper. https://ssrn.com/abstract=4284813

Adhikari, B., Alm, J., & Harris, T. F. (2021). Small business tax compliance under third-party reporting. *Journal of Public Economics*, *203*, Article 104514.

Allen, R., Hermanson, D., Kozloski, T., & Ramsay, R. (2006). Auditor risk assessment: Insights from the academic literature. *Accounting Horizons*, *20*(2), 157–177.

Atwood, T., Drake, M., & Myers, L. (2010). Book-tax conformity, earnings persistence, and the association between earnings and future cash flows. *Journal of Accounting and Economics*, *50*(1), 111–125.

Audit Analytics. (2021). *Serving their purpose? Insights on critical and key audit matters*. https://blog.auditanalytics.com/serving-their-purpose-insights-on-critical-key-audit-matters/

Bae, G., Choi, S., Lamoreaux, P., & Lee, J. (2021). Auditors' fee premiums and low-quality internal controls. *Contemporary Accounting Research*, *38*(1), 586–620.

Bakarich, K., & Kerr, J. (2016). *Compliance costs and book-tax conformity: Evidence from audit fees and audit quality*. Working paper. https://ssrn.com/abstract=2761107

Balakrishnan, K., Blouin, J., & Guay, W. (2019). Tax aggressiveness and corporate transparency. *The Accounting Review*, *94*(1), 45–69.

Bauer, A., Fang, X., & Pittman, J. (2021). The importance of IRS enforcement to stock price crash risk: The role of CEO power and incentives. *The Accounting Review*, *96*(4), 81–109.

Bell, T. B., Doogar, R., & Solomon, I. (2008). Audit labor usage and fees under business risk auditing. *Journal of Accounting Research*, *46*(40), 729–760.

Blaufus, K., Lorenz, D., Milde, M., Peuthert, B., & Schwäbe, A. N. (2022). Negotiating with the tax auditor: Determinants of tax auditors' negotiation strategy choice and the effect on firms' tax adjustments. *Accounting, Organizations and Society*, *97*, Article 101294.

Bonner, S., & Lewis, B. (1990). Determinants of auditor expertise. *Journal of Accounting Research*, *28*(Suppl), 1–20.

Boo, E., & Sharma, D. (2008). Effect of regulatory oversight on the association between internal governance characteristics and audit fees. *Accounting & Finance*, *48*(1), 51–71.

Breuer, M., & deHaan, E. (2024). Using and interpreting fixed effects models. *Journal of Accounting Research*, *62*(4), 1183–1226.

Burke, J., Hoitash, R., Hoitash, U., & Xiao, X. (2023). The disclosure and consequences of U.S. critical audit matters. *The Accounting Review*, *98*(2), 59–95.

Carcello, J. (2012). What do investors want from the standard audit report? *The CPA Journal*, *82*(1), 22–28.

Cassell, C. A., Dearden, S. M., Rosser, D. M., & Shipman, J. E. (2022). Confirmation bias and auditor risk assessments: Archival evidence. *Auditing: A Journal of Practice & Theory*, *41*(3), 67–93.

Cassell, C. A., Drake, M. S., & Dyer, T. A. (2018). Auditor litigation risk and the number of institutional investors. *Auditing: A Journal of Practice & Theory*, *37*(3), 71–90.

Causholli, M., & Knechel, W. R. (2012). An examination of the credence attributes of an audit. *Accounting Horizons*, *26*(4), 631–656.

Cho, M., Ki, E., & Koon, S. Y. (2017). The effects of accruals quality on audit hours and audit fees. *Journal of Accounting, Auditing & Finance*, *32*(3), 372–400.

Choi, J.-H., Kim, J. B., Liu, X., & Simunic, D. (2008). Audit pricing, legal liability regimes and Big 4 premiums: Theory and cross-country evidence. *Contemporary Accounting Research*, *25*(1), 55–99.

Choi, J.-H., & Wong, T. (2007). Auditors' governance functions and legal environments: An international investigation. *Contemporary Accounting Research*, *24*(1), 13–46.

Chow, T., Pittman, J., Wang, M., & Zhao, L. (2022). *Spillover effects of clients' tax enforcement on financial statement auditors: Evidence from a discontinuity design*. Working paper, Singapore Management University.

Christensen, B., Neuman, S., & Rice, S. (2019). The loss of information associated with binary audit reports: Evidence from auditors' internal control and going concern opinions. *Contemporary Accounting Research*, *36*(3), 1461–1500.

Chyz, J., Leung, W. S. C., Li, O. Z., & Rui, O. M. (2013). Labor unions and tax aggressiveness. *Journal of Financial Economics*, *108*(3), 675–698.

Dannemiller, S., Doxey, M., Hoang, K., & Houston, R. (2025). *Unique like everyone else: Auditor herding behavior in critical audit matter implementation*. Working paper. https://ssrn.com/abstract=4244294

De Simone, L., Stomberg, B., & Williams, B. (2023). Does tax enforcement disparately affect domestic versus multinational corporations around the world? *Contemporary Accounting Research*, *40*(4), 2816–2845.

Dechow, P., & Dichev, I. (2002). The quality of accruals and earnings: The role of accrual estimation errors. *The Accounting Review*, *77*(Suppl. 1), 35–59.

DeFond, M., & Zhang, J. (2014). A review of archival auditing research. *Journal of Accounting and Economics*, *58*(2–3), 275–326.

Deloitte. (2021). *Transfer pricing controversy* (4th ed.). International Tax Review. https://www.internationaltaxreview.com/itr-special-focus/transfer-pricing-controversy-4th-edition-july-2021

Desai, M. A., & Dharmapala, D. (2006). Corporate tax avoidance and high-powered incentives. *Journal of Financial Economics*, *79*(1), 145–179.

Desai, M. A., Dyck, A., & Zingales, L. (2007). Theft and taxes. *Journal of Financial Economics*, *84*(3), 591–623.

Doyle, J., Ge, W., & McVay, S. (2007). Determinants of weaknesses in internal control over financial reporting. *Journal of Accounting and Economics*, *44*(1–2), 193–223.

Drake, K., Goldman, N., Lusch, S., & Schmidt, J. (2024). Disclosure of tax-related critical audit matters and tax-related outcomes. *Contemporary Accounting Research*, *41*(2), 719–747.

Dyck, A., & Zingales, L. (2004). Private benefits of control: An international comparison. *The Journal of Finance*, *59*(2), 479–962.

Eilifsen, A., Knechel, W. R., & Wallage, P. (2001). Application of the business risk audit model: A field study. *Accounting Horizons*, *15*(3), 193–207.

El Ghoul, S., Guedhami, O., & Pittman, J. (2011). The role of IRS monitoring in equity pricing in public firms. *Contemporary Accounting Research*, *28*(2), 643–674.

Ewert, R., & Wagenhofer, A. (2019). Effects of increasing enforcement on financial reporting quality and audit quality. *Journal of Accounting Research*, *57*(1), 121–168.

Florou, A., Morricone, S., & Pope, P. (2020). Proactive financial reporting enforcement: Audit fees and financial reporting quality effects. *The Accounting Review*, *95*(2), 167–197.

Forum on Tax Administration. (2014). *Meeting on the Forum on Tax Administration (FTA): 24 October 2014, Dublin, Ireland* [Final communiqué]. https://web-archive.oecd.org/2014-10-24/326696-fta-2014-communique.pdf

Forum on Tax Administration. (2016). *Meeting of the OECD Forum on Tax Administration (FTA): 13 May 2016, Beijing* [Final communiqué]. https://web-archive.oecd.org/2016-05-13/400641-fta-communique-2016.pdf

Francis, J. R., & Wang, D. (2008). The joint effect of investor protection and Big 4 audits on earnings quality around the world. *Contemporary Accounting Research*, *25*(1), 157–191.

Gallemore, J., & Jacob, M. (2020). Corporate tax enforcement externalities and the banking sector. *Journal of Accounting Research*, *58*(5), 1117–1159.

Goldman, N., Harris, M. K., & Omer, T. (2022). Does task-specific knowledge improve audit quality: Evidence from audits of income tax accounts. *Accounting, Organizations and Society*, *99*, Article 101320.

Gotti, G., Han, S., Higgs, J. L., & Kang, T. (2012). Managerial stock ownership, analyst coverage, and audit fee. *Journal of Accounting, Auditing & Finance*, *27*(3), 412–437.

Guedhami, O., & Pittman, J. (2008). The importance of IRS monitoring to debt pricing in private firms. *Journal of Financial Economics*, *90*(1), 38–58.

Guenther, D. A., Matsunaga, S. R., & Williams, B. M. (2017). Is tax avoidance related to firm risk? *The Accounting Review*, *92*(1), 115–136.

Gutierrez, E., Minutti-Meza, M., Tatum, K. W., & Vulcheva, M. (2018). Consequences of adopting an expanded auditor's report in the United Kingdom. *Review of Accounting Studies*, *23*(4), 1543–1587.

Hanlon, M., Hoopes, J., & Shroff, N. (2014). The effect of tax authority monitoring and enforcement on financial reporting quality. *The Journal of the American Taxation Association*, *36*(2), 137–170.

Hanlon, M., Laplante, S., & Shevlin, T. (2005). Evidence for the possible information loss of conforming book income and taxable income. *Journal of Law and Economics*, *48*(2), 407–442.

HM Revenue and Customs. (2016). *Corporation tax statistics: Analyses of corporation tax receipts and liabilities, bank levy and bank surcharge*. Government of the United Kingdom. https://www.gov.uk/government/statistics/corporation-tax-statistics-2016

Hogan, C., & Wilkins, M. (2008). Evidence on the audit risk model: Do auditors increase audit fees in the presence of internal control deficiencies? *Contemporary Accounting Research*, *25*(1), 219–242.

Hoopes, J. L., Mescall, D., & Pittman, J. A. (2012). Do IRS audits deter corporate tax avoidance? *The Accounting Review*, *87*(5), 1603–1639.
Hope, O.-K., Hu, D., & Zhao, W. (2017). Third-party consequences of short-selling threats: The case of auditor behavior. *Journal of Accounting and Economics*, *63*(2–3), 479–498.
IAASB. (2020a). *Overall objectives of the independent auditor and the conduct of an audit in accordance with International Standards on Auditing* (International Standard on Auditing [ISA] 200). International Federation of Accountants.
IAASB. (2020b). *Identifying and assessing the risks of material misstatement* (International Standard on Auditing [ISA] 315 [Revised 2019]). International Federation of Accountants.
IAASB. (2020c). *Communicating key audit matters in the independent auditor's report* (International Standard on Auditing [ISA] 701). International Federation of Accountants.
IAASB. (2020d). *IAASB undertakes post-implementation review of its Revised Auditor Reporting Standards*. International Federation of Accountants.
Institute of Chartered Accountants in England and Wales. (2017). *The start of a conversation: The extended audit report*. https://www.icaew.com/technical/audit-and-assurance/faculty/the-extended-audit-report
Kennedy, P. (2008). *A guide to econometrics* (6th ed.). Blackwell Publishing.
Kerr, J. (2019). The value relevance of taxes: International evidence on the proxy for profitability role of tax surprise. *Journal of Accounting and Economics*, *67*(2–3), 297–305.
Klevak, J., Livnat, J., Pei, D., & Suslava, K. (2023). Critical audit matters: Possible market misinterpretation. *Auditing: A Journal of Practice & Theory*, *42*(3), 45–70.
Kothari, S. P., Leone, A. J., & Wasley, C. E. (2005). Performance matched discretionary accrual measures. *Journal of Accounting and Economics*, *39*(1), 163–197.
Lamoreaux, P. T. (2016). Does PCAOB inspection access improve audit quality? An examination of foreign firms listed in the United States. *Journal of Accounting and Economics*, *61*(2–3), 313–337.
Larcker, D., & Rusticus, T. (2010). On the use of instrumental variables in accounting research. *Journal of Accounting and Economics*, *49*(3), 186–205.
Lederman, L. (2010). Reducing information gaps to reduce the tax gap: When is information reporting warranted? *Fordham Law Review*, *78*(4), 1733–1760.
Lennox, C. S., Schmidt, J. J., & Thompson, A. M. (2023). Why are expanded audit reports not informative to investors? Evidence from the United Kingdom. *Review of Accounting Studies*, *28*(2), 497–532.
Levin, C. (2013). *Opening statement of Sen. Carl Levin: "Offshore profit shifting and the U.S. Tax Code, Part 2 (Apple Inc.)"* [Press release]. https://www.hsgac.senate.gov/wp-content/uploads/imo/media/doc/OPENING%20STMT%20-%20LEVIN-Carl.pdf
Li, Y., Luo, B., Singer, Z., & Zhang, J. (2025). Can critical audit matters be a signal of poor accruals quality? *Auditing: A Journal of Practice & Theory*, *44*(1), 163–187.
Liao, L., Minutti-Meza, M., Sulcaj, V., Zhang, Y., & Zou, Y. (2024). *Consequences of key audit matters in emerging economies: Evidence from Hong Kong and mainland China*. Working paper. https://ssrn.com/abstract=4776994
Lim, C. Y., Ding, D. K., & Charoenwong, C. (2013). Non-audit fees, institutional monitoring, and audit quality. *Review of Quantitative Finance and Accounting*, *41*, 343–384.
Lynch, D., Mandell, A., & Rousseau, L. M. (2024). The consequences of expanded audit reporting: Implications of tax key audit matters for tax attribute valuation and auditor-provided tax services. Working paper. https://ssrn.com/abstract=3689349
McGuire, S., Omer, T., & Wang, D. (2012). Tax avoidances: Does tax-specific industry expertise make a difference? *The Accounting Review*, *87*(3), 975–1003.
McMullin, J. L., & Schonberger, B. (2020). Entropy-balanced accruals. *Review of Accounting Studies*, *25*(1), 84–119.
McMullin, J. L., & Schonberger, B. (2022). When good balance goes bad: A discussion of common pitfalls when using entropy balancing. *Journal of Financial Reporting*, *7*(1), 167–196.
Minutti-Meza, M. (2021). The art of conversation: The expanded audit report. *Accounting and Business Research*, *51*(5), 548–581.
Nessa, M., Schwab, C., Stomberg, B., & Towery, E. (2020). How do IRS resources affect the corporate audit process? *The Accounting Review*, *95*(2), 311–338.
Niemi, L. (2002). Do firms pay for audit risk? Evidence on risk premiums in audit fees after direct control for audit effort. *International Journal of Auditing*, *6*(1), 37–51.
Nylen, J., Wangerin, D., & Zehms, K. (2024). *Differences in key audit matter and critical audit matter disclosures for the same client*. Working paper. https://ssrn.com/abstract=4569474
OECD. (2015). *Taxation administration 2015: Comparative information on OECD and other advanced and emerging economies*. https://www.oecd-ilibrary.org/taxation/tax-administration-2015_tax_admin-2015-en
OECD. (2017a). *Taxation administration 2017: Comparative information on OECD and other advanced and emerging economies*. https://www.oecd-ilibrary.org/taxation/tax-administration-2017_tax_admin-2017-en
OECD. (2017b). *The changing tax compliance environment and the role of audit*. https://www.oecd.org/en/publications/2017/09/the-changing-tax-compliance-environment-and-the-role-of-audit_g1g82ac9.html

OECD. (2019). *Taxation administration 2019: Comparative information on OECD and other advanced and emerging economies*. https://www.oecd-ilibrary.org/taxation/tax-administration-2019_74d162b6-en

OECD. (2021). *Taxation administration 2021: Comparative information on OECD and other advanced and emerging economies*. https://www.oecd-ilibrary.org/taxation/tax-administration-2021_cef472b9-en

O'Keefe, T., Simunic, D., & Stein, M. (1994). The production of audit services: Evidence from a major public accounting firm. *Journal of Accounting Research*, *32*(2), 241–261.

PCAOB. (2017a). *Audit risk* (Auditing Standard 1101).

PCAOB. (2017b). *Identifying and assessing risks of material misstatement* (Auditing Standard 2110).

PCAOB. (2017c). *The auditor's report on an audit of financial statements when the auditor expresses an unqualified opinion* (Auditing Standard 3101).

PCAOB. (2022). *Post-implementation review of auditor reporting*. https://pcaobus.org/oversight/standards/pir/auditor-reporting

Pinto, I., & Morais, A. (2019). What matters in disclosures of key audit matters: Evidence from Europe. *Journal of International Financial Management & Accounting*, *30*, 145–162.

Pomeranz, D. (2015). No taxation without information: Deterrence and self-enforcement in the value added tax. *American Economic Review*, *105*(8), 2539–2569.

Qi, Y., Roth, L., & Wald, J. (2010). Political rights and the cost of debt. *Journal of Financial Economics*, *95*(2), 202–226.

Rousseau, L., & Zehms, K. (2024). It's a matter of style: The role of audit firms and audit partners in key audit matter reporting. *Contemporary Accounting Research*, *41*(1), 529–561.

Russett, B. M. (1969). Who pays for defense? *American Political Science Review*, *63*(2), 412–426.

Shepardson, M. L. (2019). Effects of individual task-specific experience in audit committee oversight of financial reporting outcomes. *Accounting, Organizations and Society*, *74*, 56–74.

Shroff, N., Verdi, R., & Yu, G. (2014). Information environment and the investment decisions of multinational corporations. *The Accounting Review*, *89*(2), 759–790.

Sierra-Garcia, L., Gambetta, N., Garcia-Benau, M., & Orta-Perez, M. (2019). Understanding the determinants of the magnitude of entity-level risk and account level risk key audit matters: The case of the United Kingdom. *The British Accounting Review*, *51*(3), 227–240.

Slemrod, J., Collins, B., Hoopes, J. L., Reck, D., & Sebastiani, M. (2017). Does credit-card information reporting improve small-business tax compliance? *Journal of Public Economics*, *149*, 1–19.

Smith, K. W. (2023). Tell me more: A content analysis of expanded auditor reporting in the United Kingdom. *Accounting, Organizations and Society*, *108*, Article 101456.

Smith, R. P. (1977). Military expenditure and capitalism. *Cambridge Journal of Economics*, *1*(1), 61–76.

Sulcaj, V. (2023). *Litigation risk, financial reporting quality, and critical audit matters in the audit report: Early U.S. evidence*. Working paper. https://ssrn.com/abstract=3771081

Taylor, M. (2010). The effects of industry specialization on auditors' inherent risk assessments and confidence judgements. *Contemporary Accounting Research*, *17*(4), 693–712.

Thibodeau, J. C. (2003). The development and transferability of task knowledge. *Auditing: A Journal of Practice and Theory*, *22*(1), 47–67.

Trotman, K., & Wright, W. (2012). Triangulation of audit evidence in fraud risk assessments. *Accounting, Organizations and Society*, *37*(1), 41–53.

Vogele, A., & Bader, W. (2001). German electronic audit looms. *International Tax Review*, *12*(3), 19–25.

Yost, B., & Shu, S. (2022). Does tax enforcement deter managers' self-dealing? *Journal of Accounting and Economics*, *74*(1), Article 101512.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Filosa, J. R., Huang, J. (J.), Lei, L. (G.), & Stein, S. E. (2025). Does tax enforcement inform auditors' risk assessment? Evidence from key audit matters. *Contemporary Accounting Research*, *42*(2), 1423–1454. https://doi.org/10.1111/1911-3846.13042

APPENDIX: VARIABLE DEFINITIONS

Variable	Definition (Source)
NKAM	Number of KAMs disclosed in the auditor's report of company *i* in year *t* (Audit Analytics)
NTAXKAM	Number of tax-related KAMs disclosed in the auditor's report of company *i* in year *t*. Tax-related KAMs refer to those classified as one of the following Audit Analytics topics: other income taxes (27), deferred income taxes (28), and uncertain tax positions (29) (Audit Analytics)
NOTHERKAM	Number of nontax KAMs disclosed in the auditor's report of company *i* in year *t* (Audit Analytics)
TAXENF	FTEs within the "audit, investigation, and other verification" function of the tax authority for country *j* in year *t* scaled by inflation-adjusted GDP in year *t* (in billions of dollars). FTE of 1.0 indicates resources equal to one staff member available for one full year. GDP is in 2019 constant dollars. For ease of interpretation of the regression coefficients, we divide the ratio by 10 (ISORA)
TAXENF_DUM	Indicator variable that equals one if *TAXENF* is above the sample median, and zero otherwise
CONFORMITY	Book-tax conformity is calculated by estimating eq. 3 of Atwood et al. (2010) within each country and year. Specifically, this measure is the scaled ranking of the root mean square errors (RMSEs) obtained from a country-year regression of current tax expense on pre-tax book income and total dividends, all scaled by average total assets. The scaled ranking is multiplied by negative one so that a larger value suggests a higher level of book-tax conformity. We use the average of the country's book-tax conformity in the prior year and current year (year t and year $t-1$) in our subsample analysis (Compustat Global)
AUDITOR_TAXEXP	Exposure of the auditor to complex tax issues is estimated following a similar approach to Goldman et al. (2022). For each client, we determine whether it has R&D activities, foreign operations, or goodwill (0 if the client has none of the attributes, 1 if the client has one attribute, 2 if the client has two attributes, and 3 if the client has all three attributes). Note that our measure differs from Goldman et al. (2022) by using goodwill to capture M&A activity since tax loss carryforward data are not available for the companies in our sample. We sum the client values at the audit firm-office-year and use the average of this measure in the prior year and current year (year t and year $t-1$) in our subsample analysis (Compustat Global and Audit Analytics)
SIZE	Natural logarithm of total assets of company *i* at the end of year *t* (currency unit in US dollars) (Compustat Global)
LEVERAGE	Leverage ratio computed as the sum of long-term debt and current portion of long-term debt scaled by total assets of company *i* in year *t* (Compustat Global)
ΔSALE	Year-over-year sales growth of company *i* in year *t* (Compustat Global)
LOSS	Indicator variable that equals one if company *i* reports negative income in year *t*, and zero otherwise (Compustat Global)
FOREIGN	Indicator variable that equals one if company *i* reports foreign exchange income/loss in year *t*, and zero otherwise (Compustat Global)
AUDITFEE	Natural logarithm of one plus total audit fees of company *i* in year *t* (currency unit in US dollars) (Audit Analytics)
BIG4	Indicator variable that equals one if company *i* hires a Big 4 auditor in year *t*, and zero otherwise (Audit Analytics)
VOL_ETR	Standard deviation of the cash effective tax rate of company *i* over the past 5 years ($t-4$ to t). The cash effective tax rate of company *i* is calculated as income tax paid scaled by pre-tax income minus special items in year *t* (Compustat Global)
VOL_OCF	Standard deviation of operating cash flow of company *i* scaled by lagged total assets over the past 5 years ($t-4$ to t) (Compustat Global)

(Continues)

APPENDIX (Continued)

Variable	Definition (Source)
VOL_SPI	Standard deviation of special items scaled by lagged total assets of company *i* over the past 5 years ($t-4$ to t) (Compustat Global)
VOL_PI	Standard deviation of pre-tax income scaled by lagged total assets of company *i* over the past 5 years ($t-4$ to t) (Compustat Global)
ETR	Cash effective tax rate of company *i* calculated as income tax paid scaled by pre-tax income minus special items in year *t* adjusted by country *j*'s statutory tax rate following De Simone et al. (2023). If the value is missing, we replace it with zero to avoid sample attrition. If the value exceeds one, we replace it with one to alleviate the impact of outliers (Compustat Global)
FRQ	Financial reporting quality index of company *i* in year *t* computed as the mean of 10 minus the decile rank of the two financial reporting quality measures used in Hanlon et al. (2014) and then divided by nine. The two financial reporting quality measures are the (1) absolute value of abnormal accruals calculated following Kothari et al. (2005) and (2) absolute value of abnormal accruals calculated following Dechow and Dichev (2002). If the value is missing, we replace it with zero to avoid sample attrition. The index ranges between zero and one with a higher value indicating better reporting quality (Compustat Global)
HIGH_MILITARY	Indicator variable that equals one if the ratio of military spending to GDP for country *i* in year *t* is in the top quartile of the sample, and zero otherwise (World Bank)
ANALYST	Median number of analysts following companies in country *j* in year *t* (I/B/E/S)
FREE_PRESS	Degree of freedom available to journalists of country *j* in year *t* (in rank) (Reporters Without Borders)
REG_QUALITY	Perceptions of the ability of the government to formulate and implement sound policies and regulations that permit and promote private sector development of country *j* in year *t* (in rank) (World Bank)
TRADE_UNION	Trade union density of country *j* in year *t*, which is defined as the number of net union members (i.e., excluding those who are not in the labor force, unemployed and self-employed) as a proportion of the number of employees (OECD)
EASE_OF_CREDIT	Sum of the strength of the legal rights index and the depth of credit information index of country *j* in year *t* (in rank). The World Bank's Doing Business measures the legal rights of borrowers and lenders with respect to secured transactions through one set of indicators and the reporting of credit information through another. The first set of indicators measures whether certain features that facilitate lending exist within the applicable collateral and bankruptcy laws. The second set of indicators measures the coverage, scope, and accessibility of credit information available through credit reporting service providers, such as credit bureaus or credit registries (World Bank)
VENTURE_CAP	Number of venture capital deals in country *j* per billion GDP in US dollars based on purchasing power parity (PPP) in year *t* (in rank) (World Bank)